

System requirements

To-Do *Space*

Classification	Productivity Application
Status	Beginning of the implementation phase
Program name	To-Do Space
Project number	1
Project manager	Project manager
Version	0.3
Date	20/12/2022
Project sponsor	/
Author(s)	Elise Høimyr
Distribution	/

List of changes

Version	Date	Change	Author
0.1	22/11/2022		
0.2	12/12/2022	New refined Use Case diagrams and sequence diagrams. Precision added to the attributes and operations of the class diagrams and details to the sequence diagrams. Encryption algorithm preference in NFRs changed. Added table for main and alternate flows.	Elise Høimyr
0.3	19/12/2022	New Use case diagram of all functional requirements, split in several parts for more clarity. Added synchronization logic to the 3 sequence diagrams. Added: -New class diagrams for model, view and control. -Concept of the application. -Screenshots of the prototype. -Index with list of features and list of diagrams	Elise Høimyr

Table 1: Change control

Description

The system requirements describe requirements for the future system. They are structured according to requirement categories. They include business requirements, operating requirements, support requirements and security requirements, for example, and they are prioritized according to their importance. The documentation of the system requirements is based on and uses the standards/notations of the requirement engineering method used.

Source: Hermes

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1. System overview

1.1. Concept

To do *Space* values the ideas of their users above all: as if they were stars in the giant space that is the user's mind. However, stars alone difficulty manage to shine bright to distant planets.

As such, the aim of To do *Space* is to allow the user to easily create the constellations that are projects, the galaxies that are teams, with each one of their stars, the tasks. In other words; making the most out of their ideas.

To do so the application aims at a highly customizable model, with many features allowing a lot of flexibility to the users. Although it lends more on the functionality than on the simplicity side, it remains an application that should be user-intuitive with its graphical design and the presence of an interactive tutorial that shall guide the user throughout their entire experience with To do *Space*.

With many unique features that we never see on a single application but rather as widespread on many but different ones, To do *Space* should not just be a To-Do app but a true planning workspace that may serve users working in any field, whether it may be for personal or professional use.

Of course, the application is designed to encourage team work as well.

These objectives should be achieved by structuring the application around an account based system where files that are organized just like those that lie on any physical computer: Stored in folders, and accessible through a path. However, here the files are the tasks. Additionally, the storage space for each account is located on a cloud, which allows transferring the content between devices, a crucial feature for teamwork.

All the additional features are either implemented on the tasks or themselves, or on the display screen.

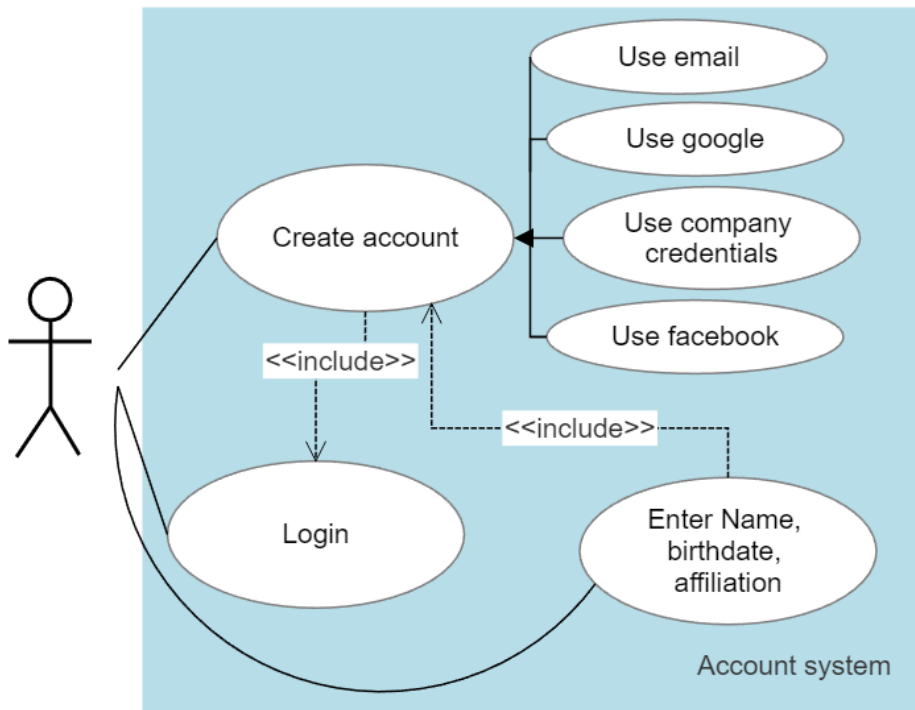
The tasks can indeed support external programs as features: a text editor like Microsoft Word or Google Docs for example could be implemented to write the descriptions of the tasks, etc.

Another example: relying on an external 3D view application would be the most convenient solution to give the option to display the tasks as 3D objects on the projects page.

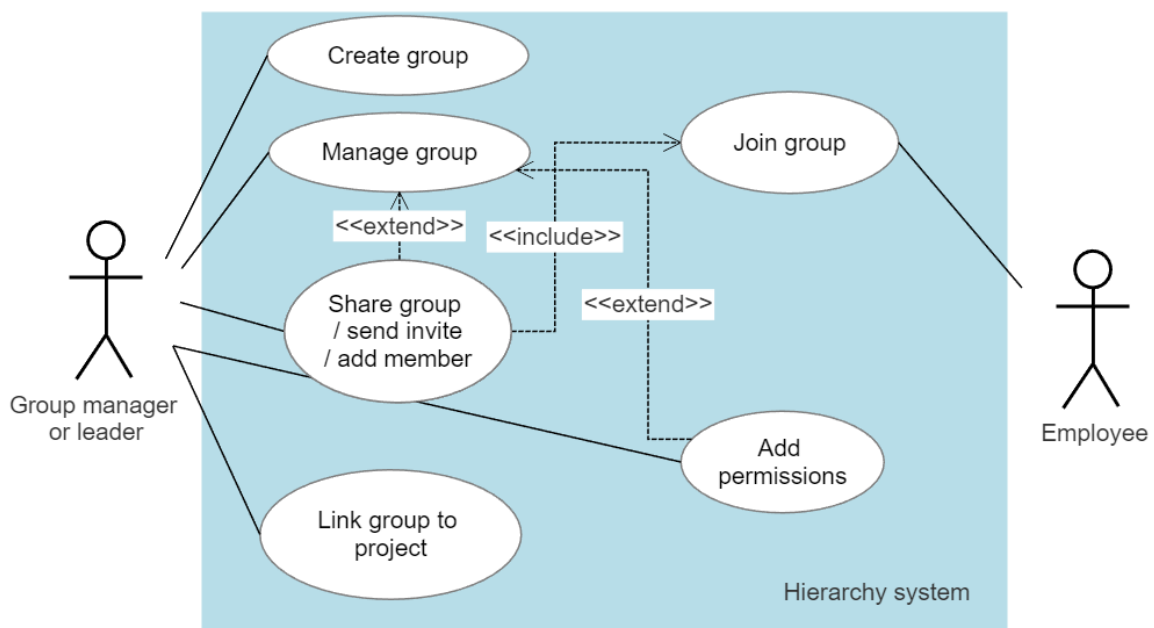
In the end, these innovative features are just either variations of classical ones we know (represented by generalizations on the uml diagrams), or additions to a simple to-do app inspired from other types of systems. These each considerably increase the number of possibilities and get the application closer to the ideal of a boundaryless space to organize the user's ideas.

1.2: UML Use case diagrams of most functional requirements:

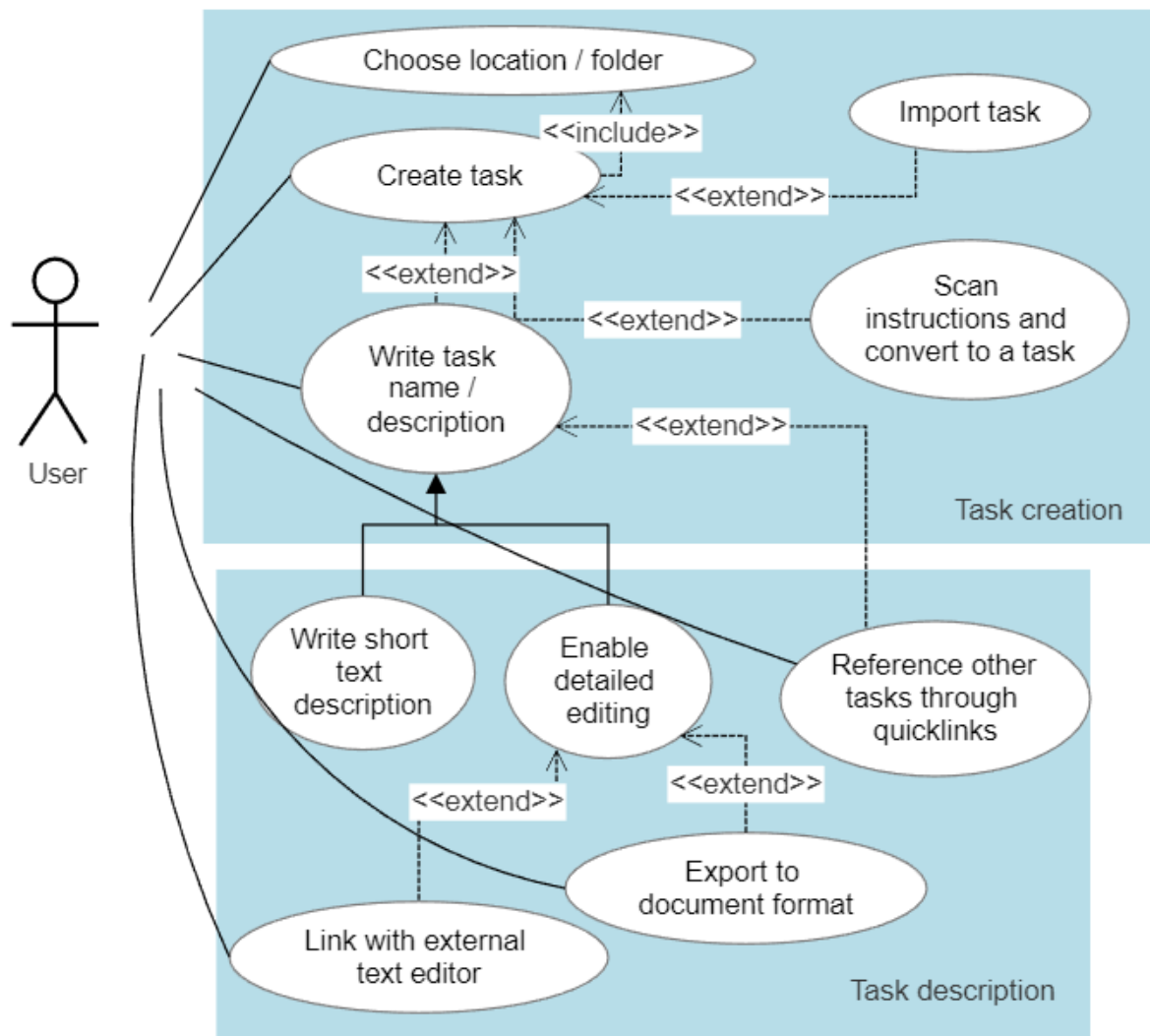
Account system:



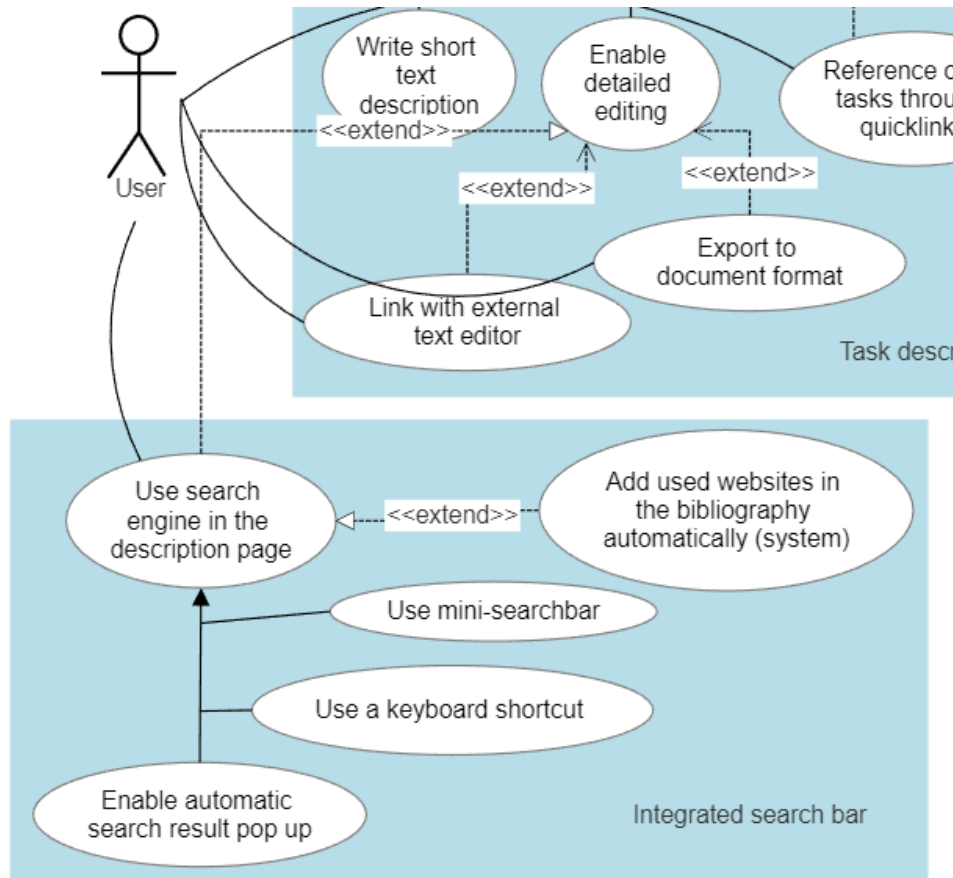
Hierarchy system:



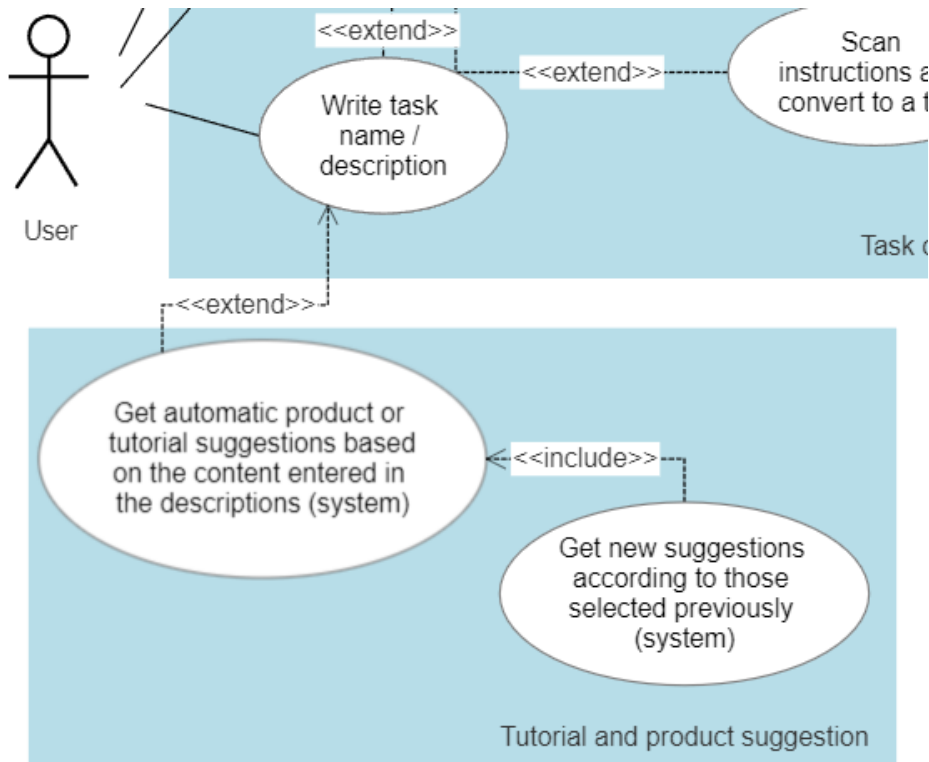
Task creation and description:



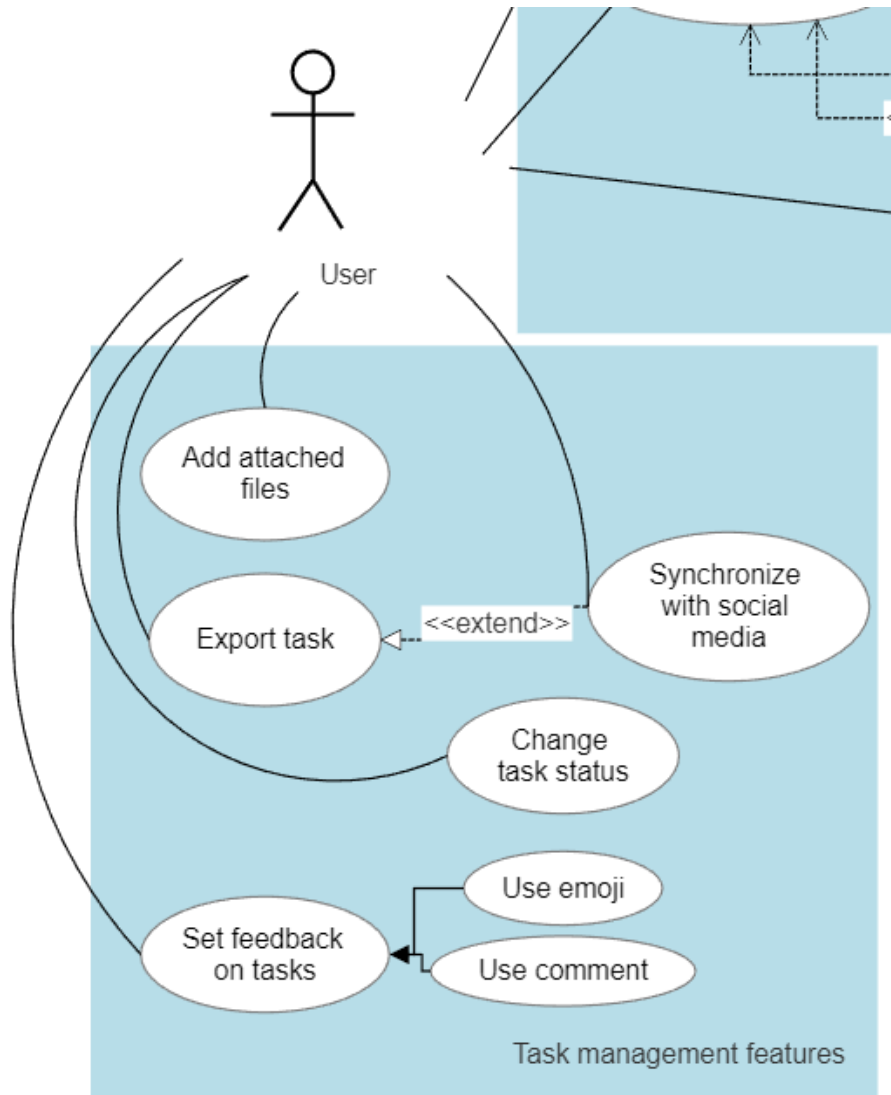
Integrated search bar (linked to description):



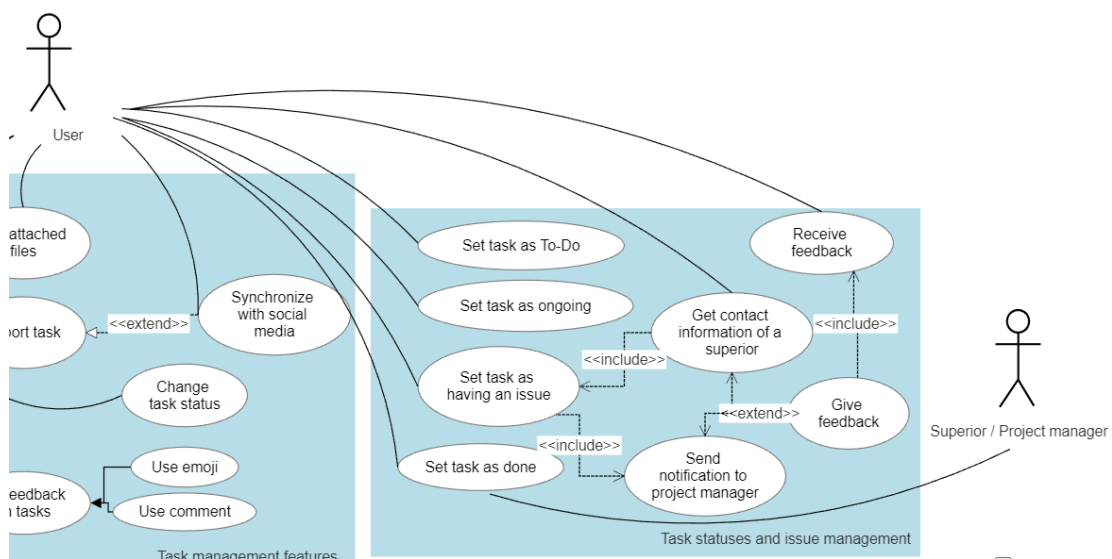
Tutorial and product suggestion (linked to the description):



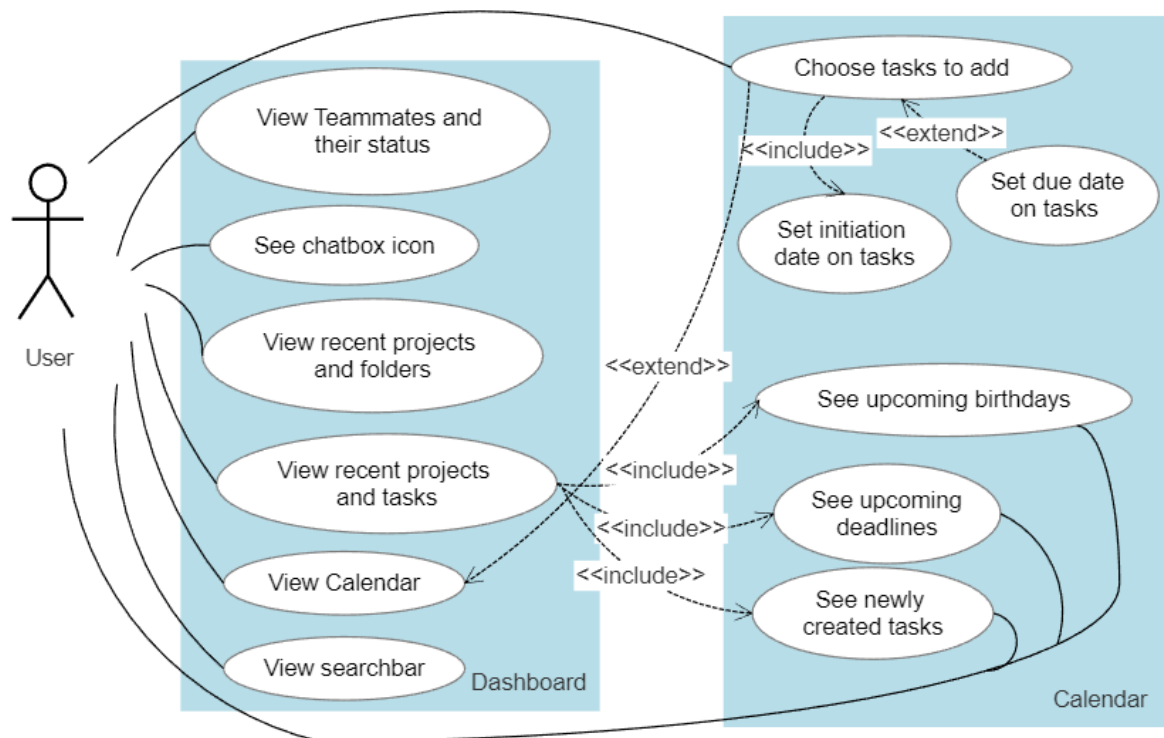
Task management features:



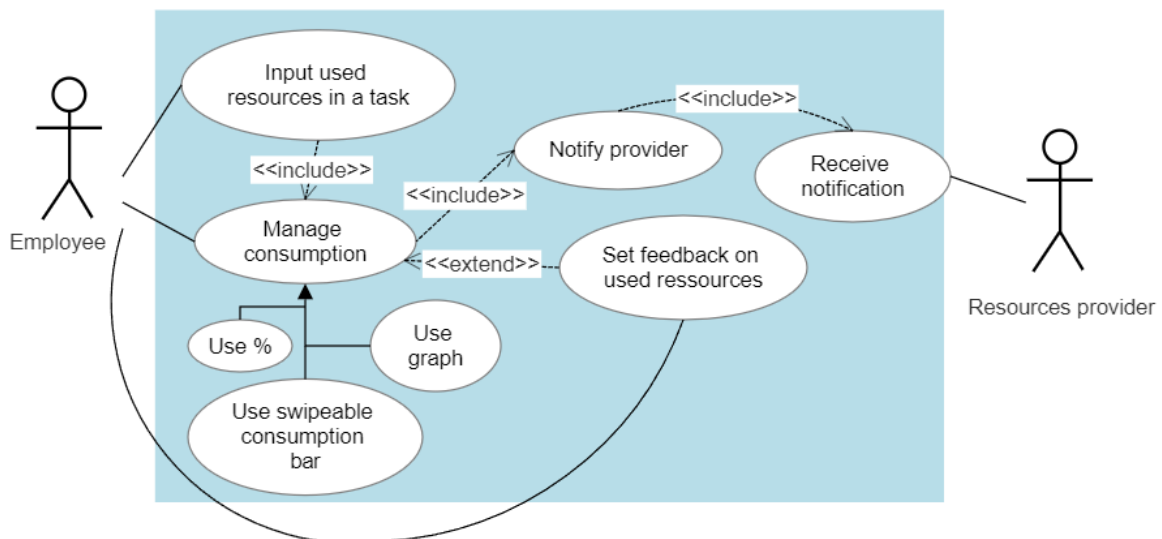
Task status (linked to task management features):



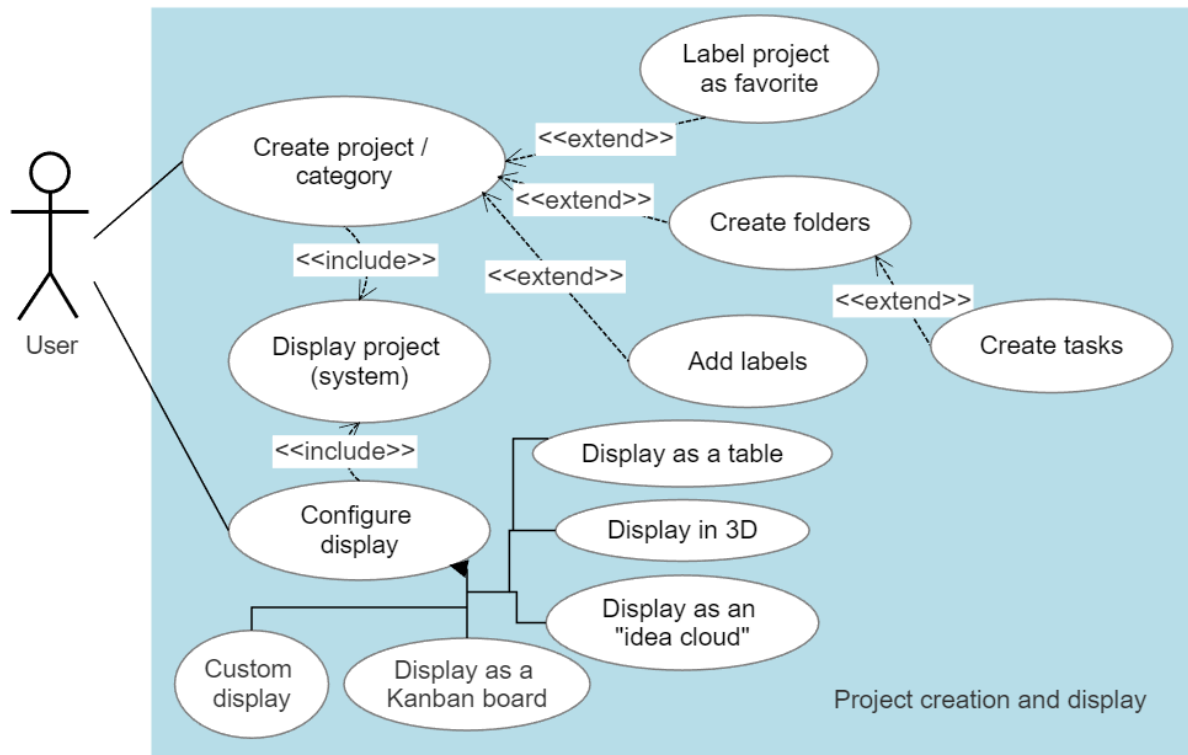
Dashboard + Calendar:



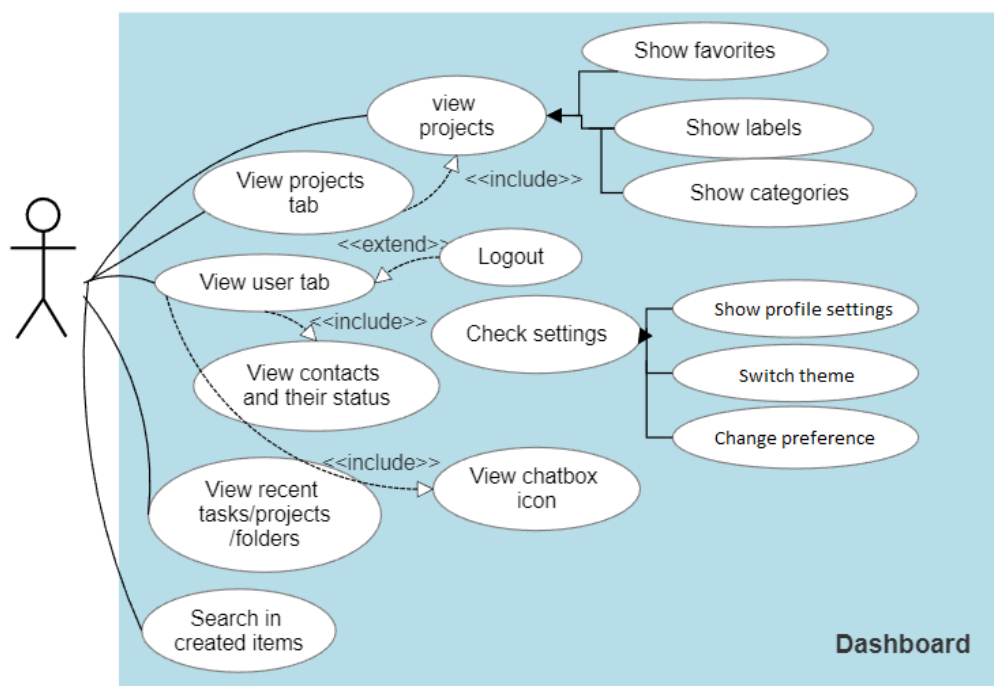
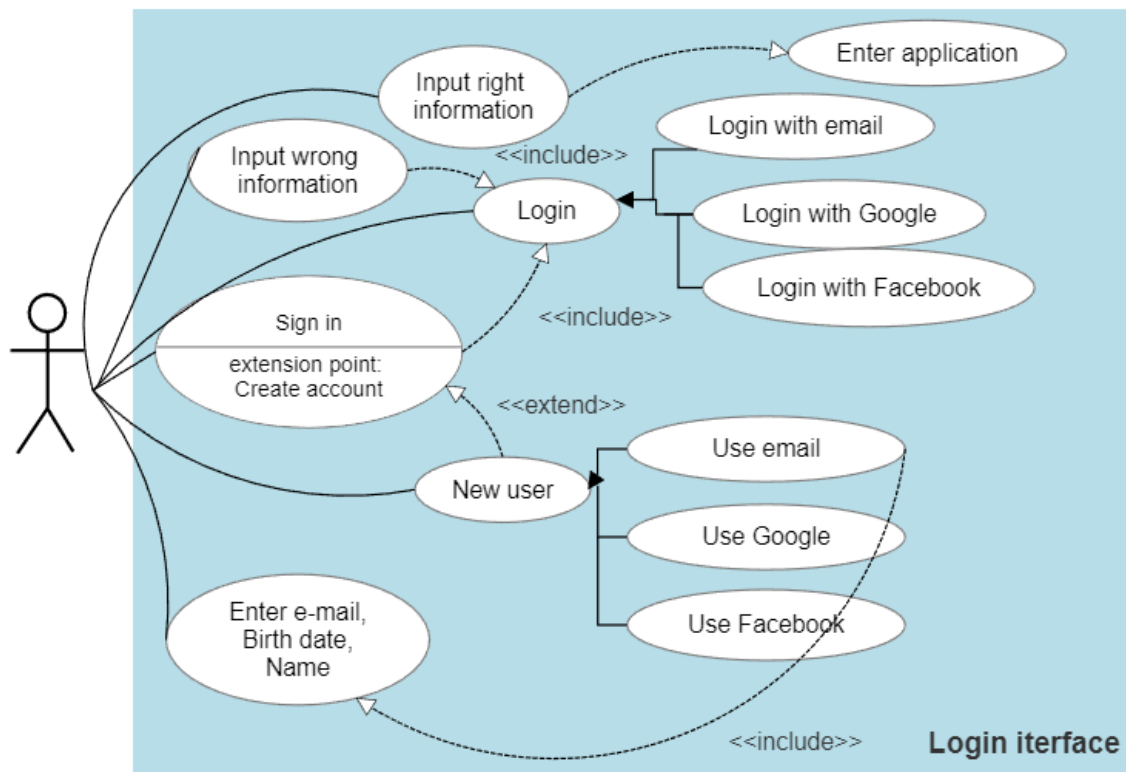
Project resources management:

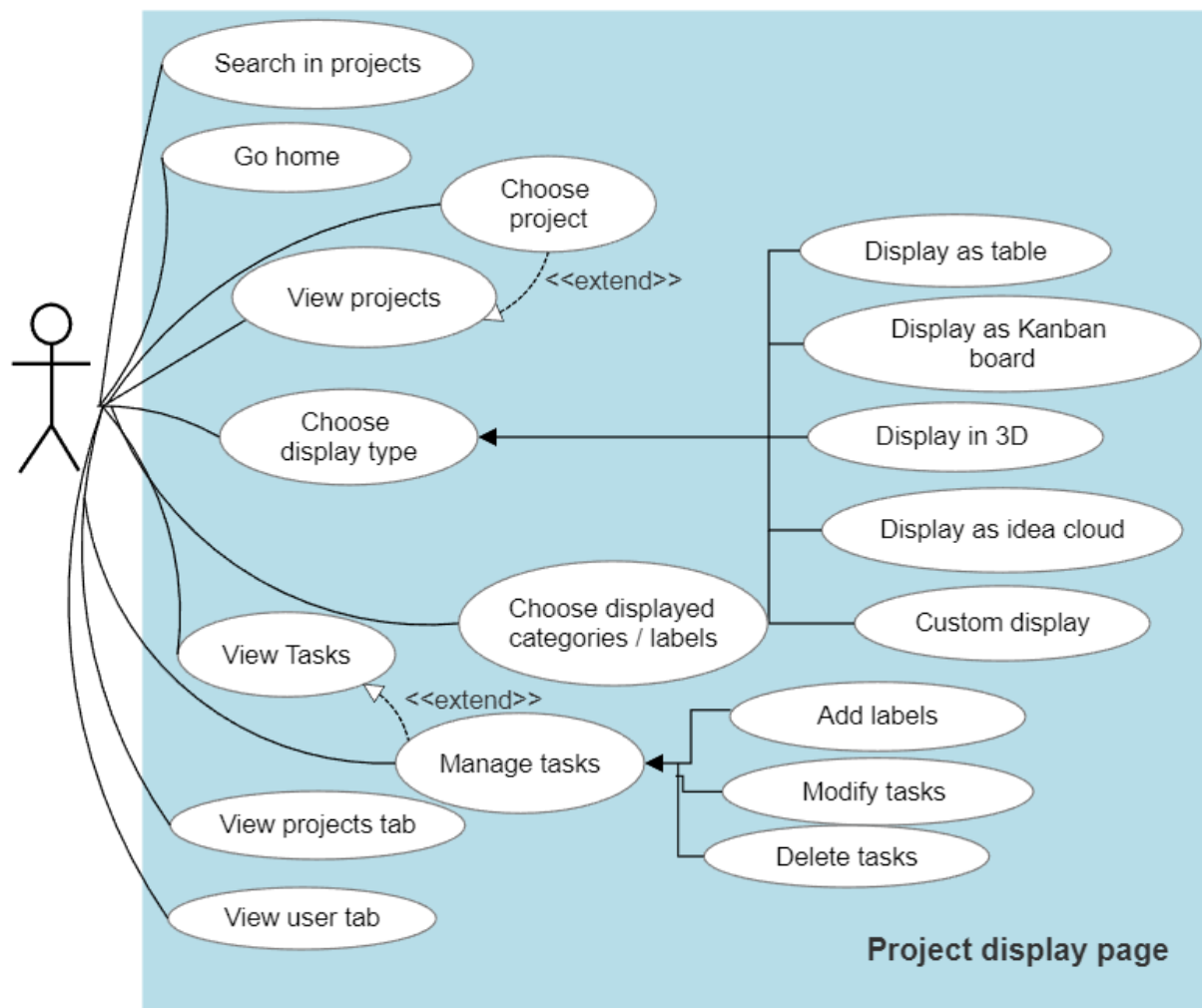


Project display:

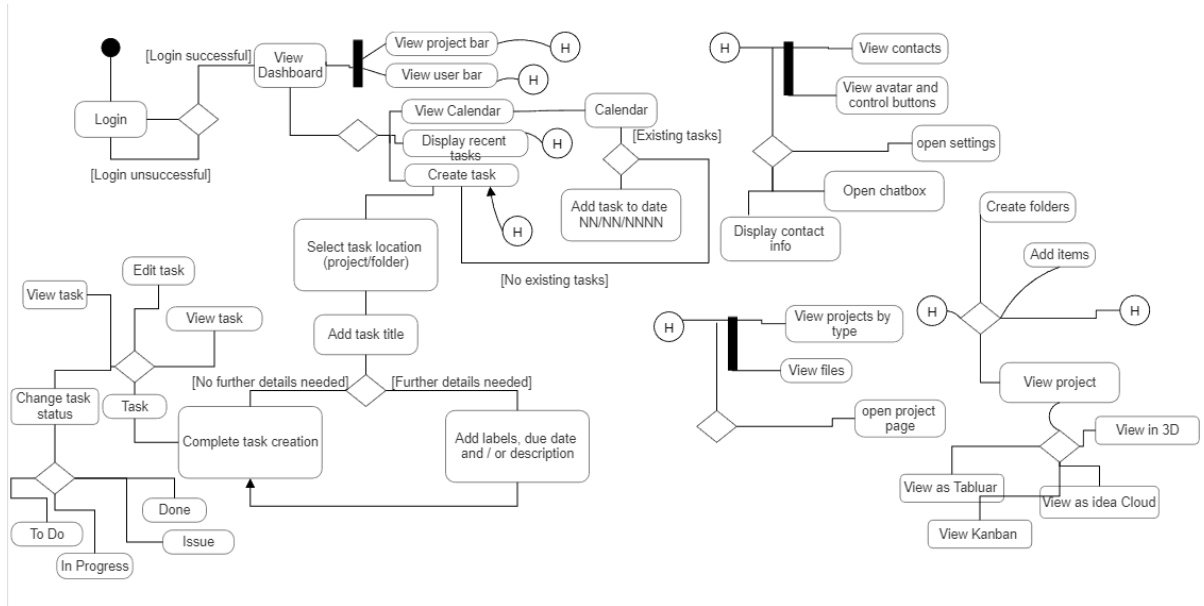


1.3: Fully described use cases of some main screens:



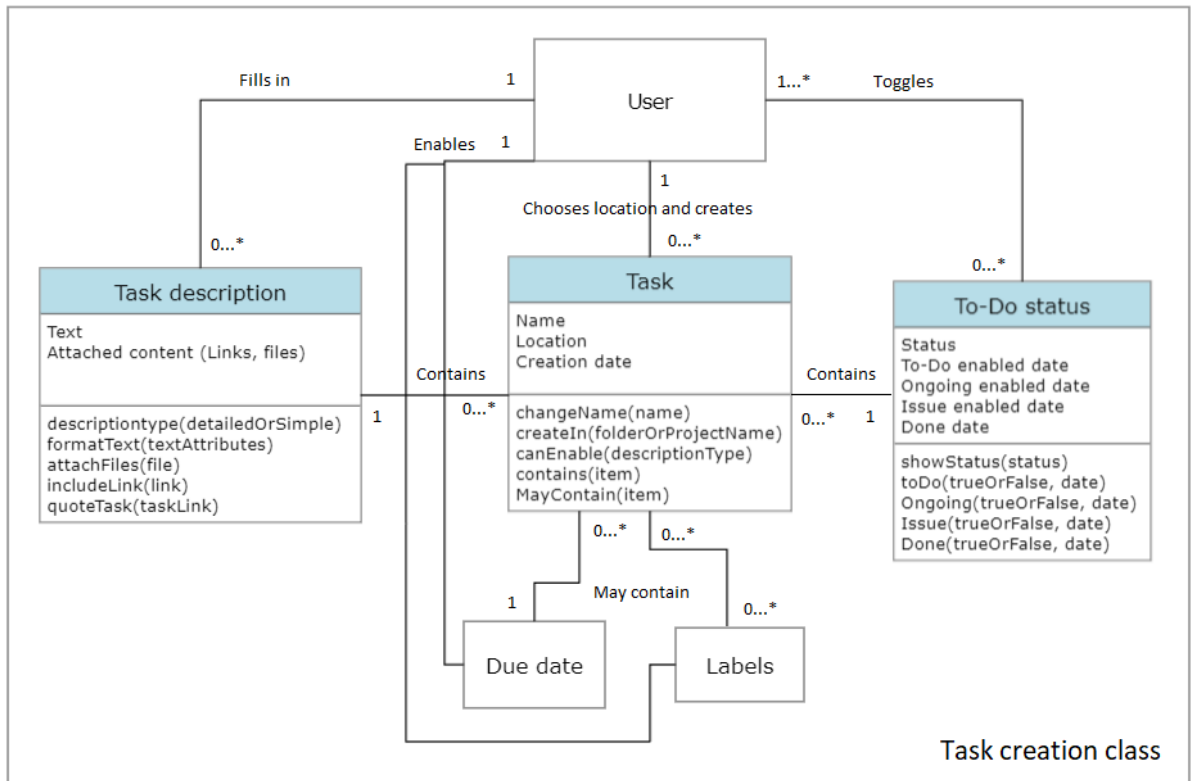


1.4 Activity diagram of the main screens:

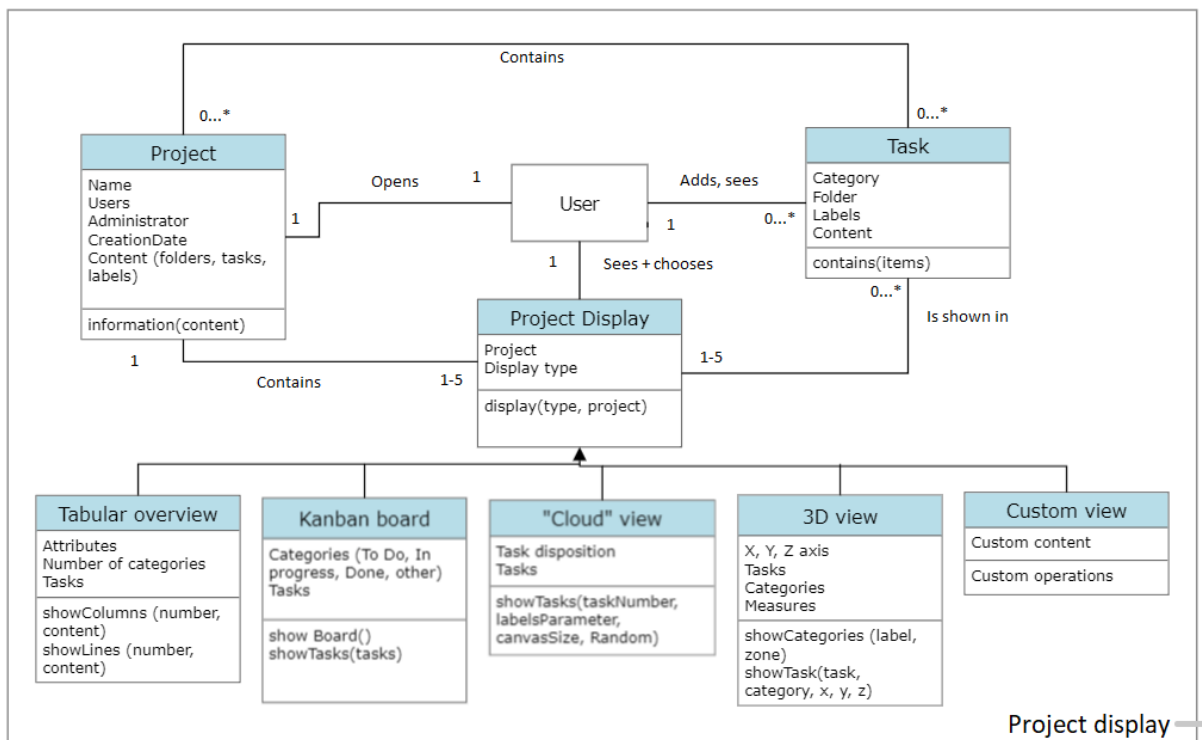


1.5.1 UML class diagrams of some main features, and two innovative ones:

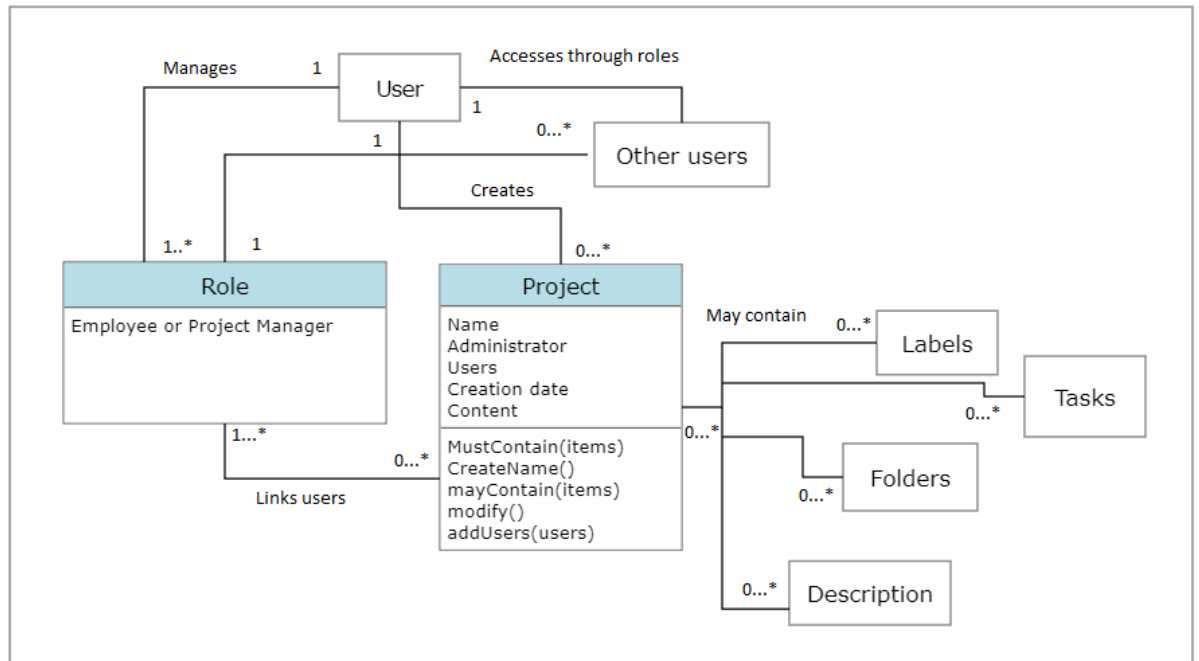
Task creation:



Project and task display:



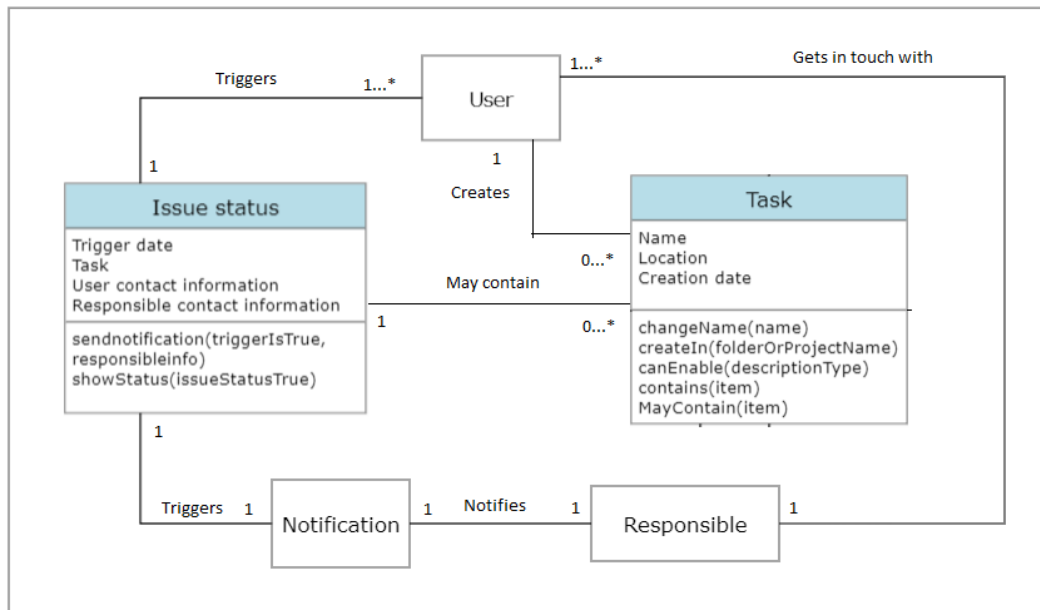
Project creation:



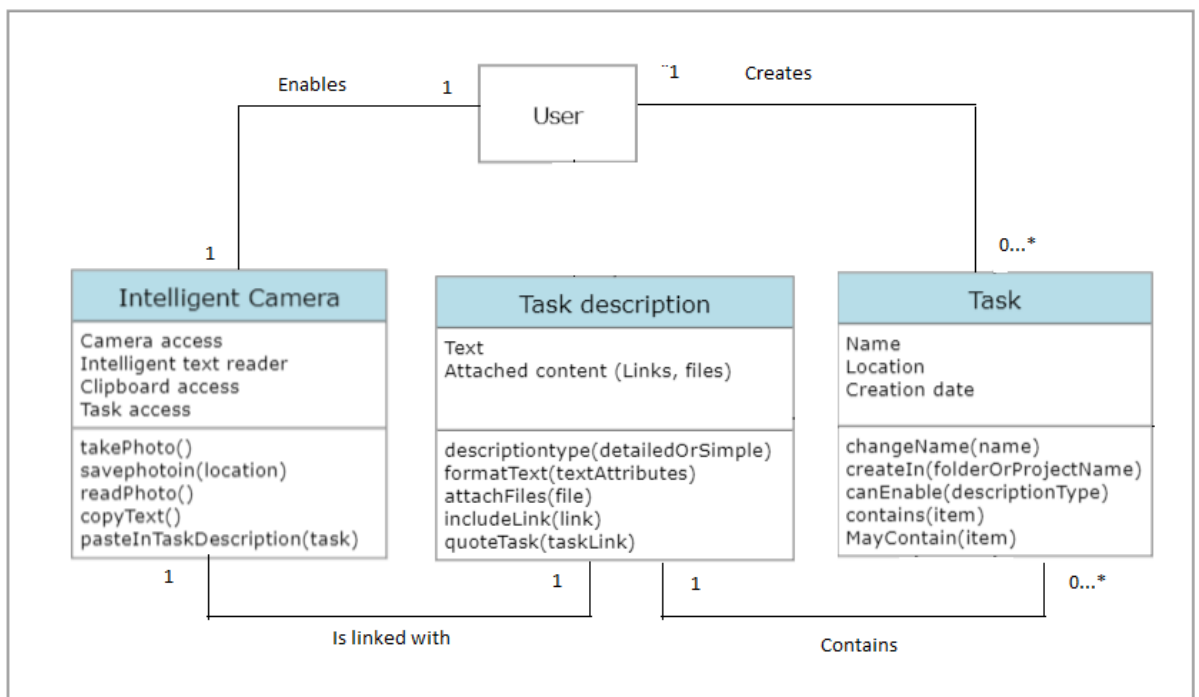
Label creation:

Create label
Project/folder location Color Name User
assignName(name) changeColor(color) addtoproject(projectID) addtotask(taskID)

Task issue status:

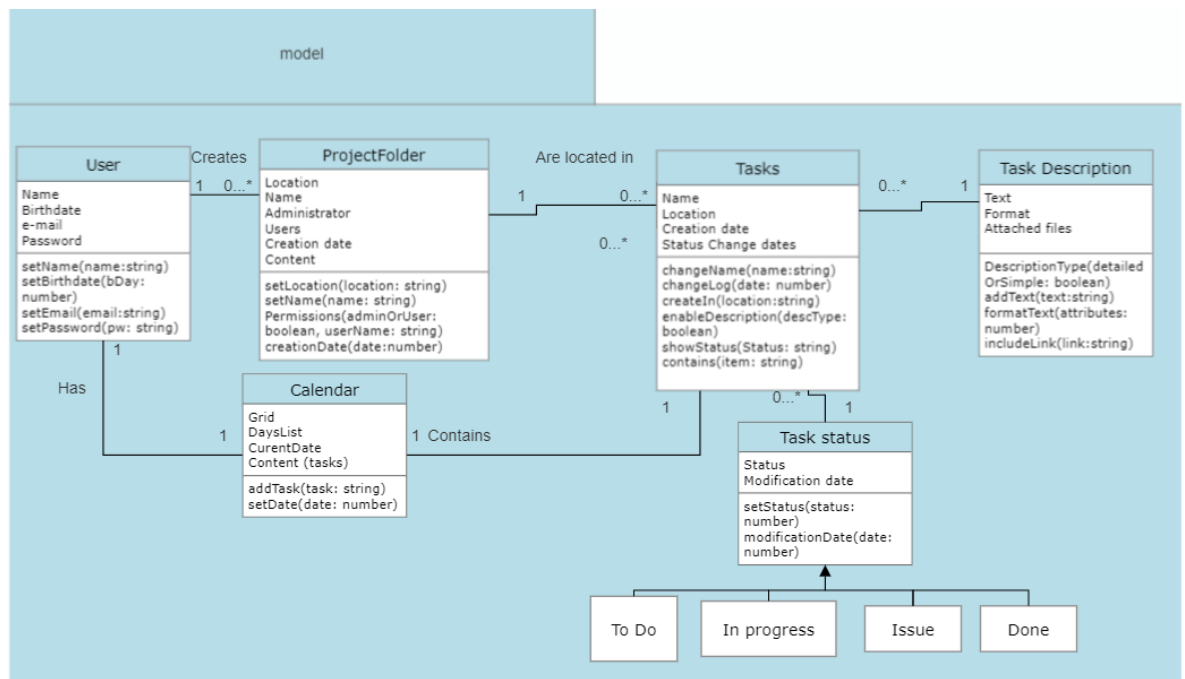


Task camera scan feature:

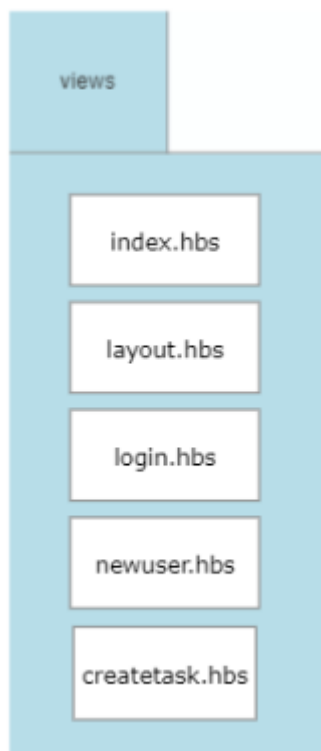


1.5.2 UML Class diagrams of model, view and control:

Model:

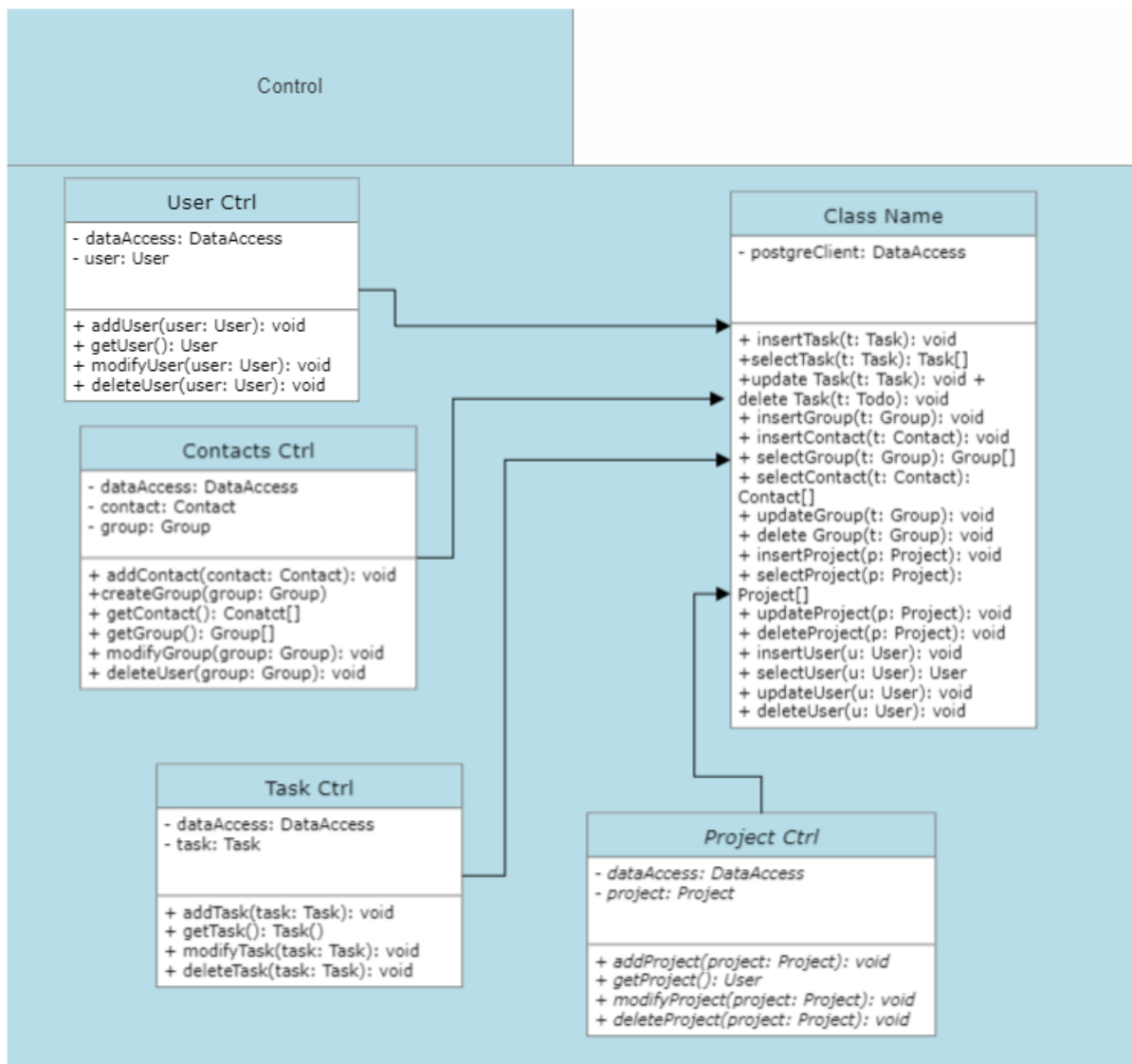


Views:



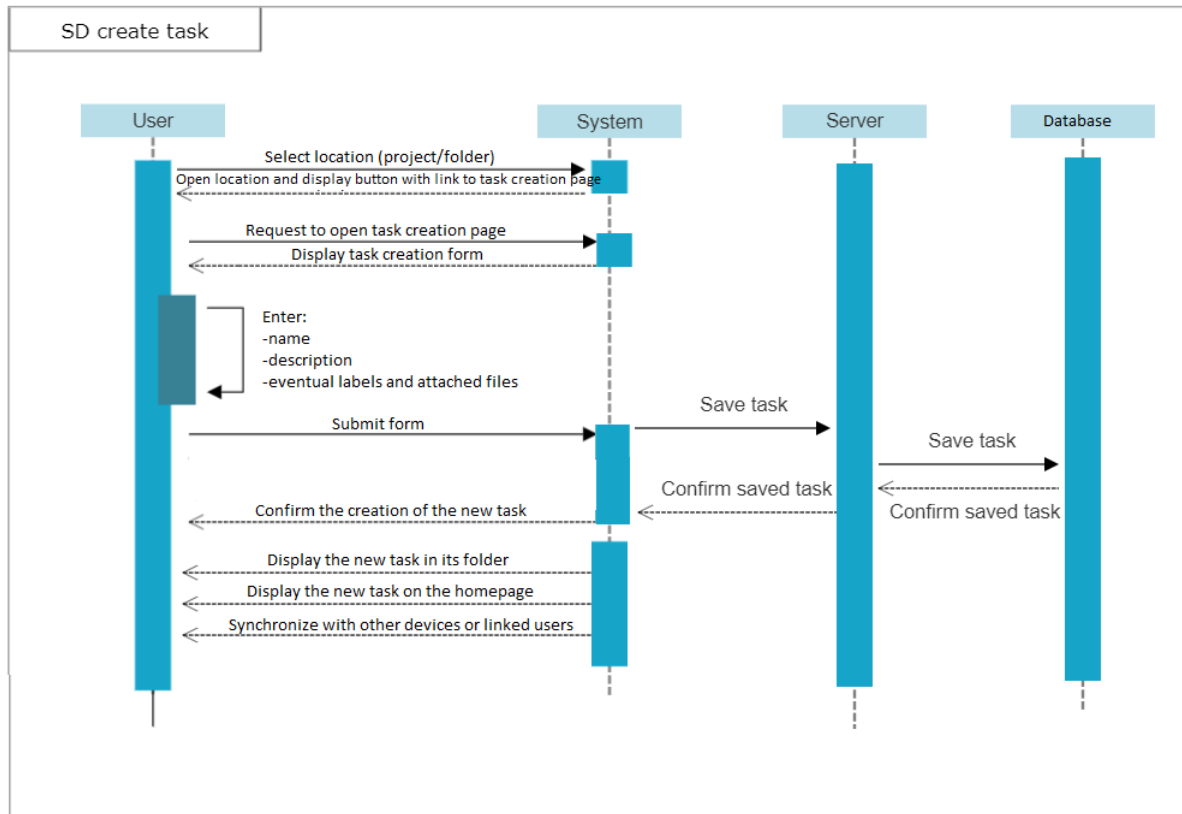
System requirements
To Do Space

Control:

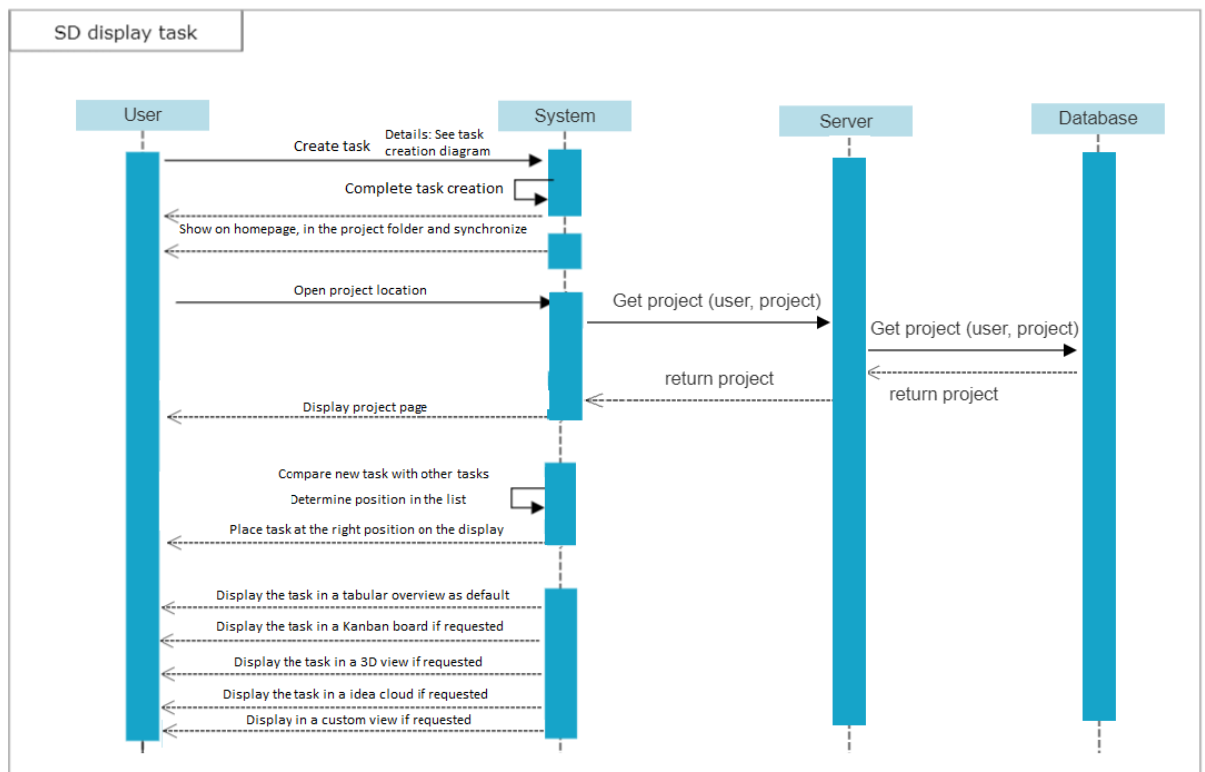


1.6 UML sequence diagrams for task creation, task synchronization and task display:

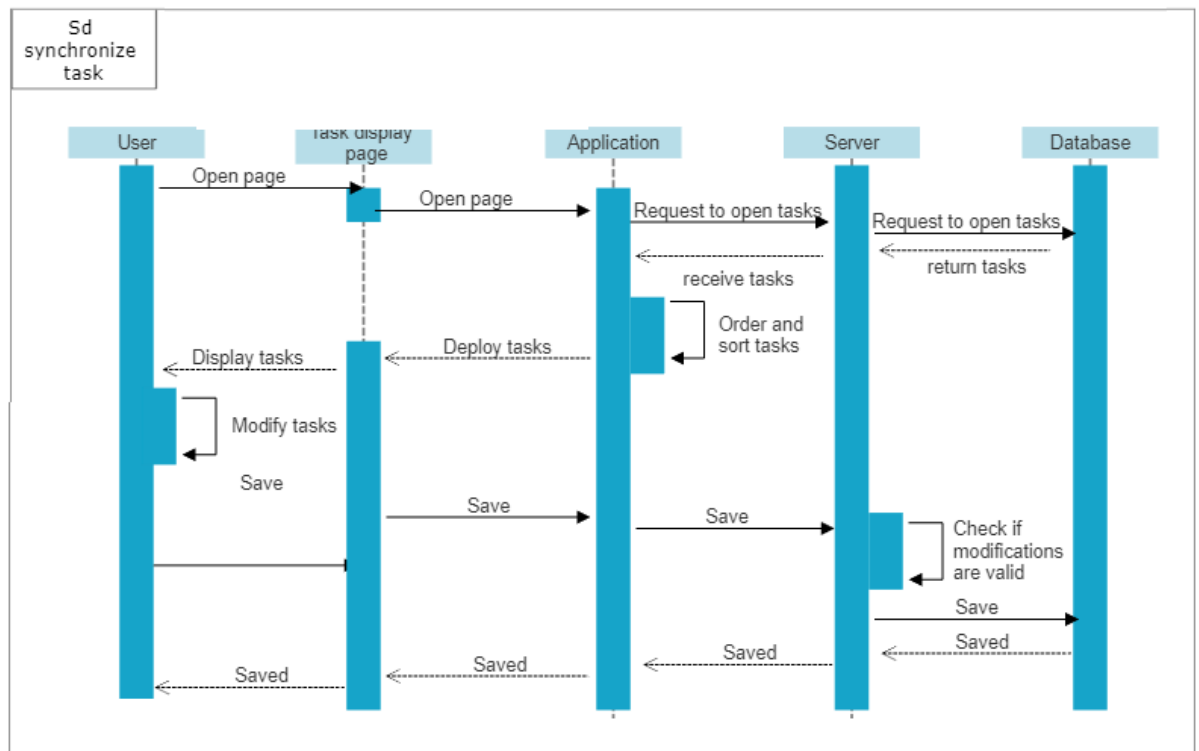
Create task:



View task:



Synchronize task:



1.7. Tabular description of the main and alternate flows for the main requirements
(main requirements: see activity diagram)

Use case	Description
Login	1. Brief Description Login process that is necessary for any user to access the content of the application. 2. Actors User 2.1. Systems To-Do Space system Google Facebook 3. Flow of Events • <u>3.1. Main (basic) Flow</u> → The USER enters his username and password → The system verifies the authentication and approves the login → The system allows the USER to enter the application and displays the home page • <u>3.2. Alternate Flows</u> 3.2.1. Login with Facebook → The USER clicks on the login with Facebook button → The system asks the user for permission to get access to their personal information provided on Facebook that is necessary for the login → The system requests the information from Facebook → Facebook provides the credentials → The system approves the login → The system allows the USER to enter the application and displays the home page 3.2.2 Login with Google → The USER clicks on the login with Google button → The system asks the user for permission to get access to their personal information provided on Google that is necessary for the login → The system requests the information from Google → Google provides the credentials → The system approves the login → The system allows the USER to enter the application and displays the home page 3.2.3 Login unsuccessful → The USER enters invalid credentials

	→ The system disapproves the login and displays an error message → The system saves the date and location of the unsuccessful login attempt, and removes one of the 10 chances to enter the right password 3.2.4 Repeated unsuccessful logins → The user burns all chances to provide the right password → The system blocks the login and displays an error message, sends an e-mail to the owner of the account 4. Special Requirements Facebook login and Google login available. 5. Pre-conditions No active session, only the To-Do Space home and advertisement pages are available, as well as the login page. 6. Post-conditions User session open and application content unlocked. 7. Extension Points Create account
View dashboard	1. Brief Description Main page of the user interface displayed by default after login. Contains access links to the other parts of the application. 2. Actors User 3. Flow of Events • <u>3.1. Main (basic) Flow</u> → The system displays a minimized User bar → The system displays a minimized Project bar → The system displays a Task creation button, the Calendar, and Recent tasks • <u>3.2. Alternate Flows</u> 3.2.1 Open User bar → In addition the the items displayed in the main flow, the system displays the user's contacts and their status, the user's profile picture, a chatbox button and a shortcut to the settings page 3.2.2 Open Project bar → In addition the the items displayed in the main flow, the system displays shortcuts to the created categories, favorite projects, and labels and a list of the projects. 4. Special Requirements / 5. Pre-conditions

	Other or no displayed page 6. Post-conditions Dashboard visible 7. Extension Points Calendar Project Page Settings Search page Chatbox
Use calendar	1. Brief Description Allows the USER to view upcoming tasks, place existing tasks at specific dates, and schedule events by adding new tasks 2. Actors User 3. Flow of Events • 3.1. Main (basic) Flow → The USER requests to open the calendar via the dashboard → The system displays the calendar page → The USER clicks on a time slot → The system shows the description of that time slot and suggests to add content → The USER chooses a task from the existing tasks list to add to the the time slot → The system saves the content and displays it on the calendar • 3.2. Alternate Flows 3.2.1 Create new task directly in the calendar → The USER requests to open the calendar via the dashboard → The system displays the calendar page → The USER clicks on a time slot → The system shows the description of that time slot and suggests to add content → The USER clicks on the “create event” button and is redirected to the task creation page → The system saves the content and displays it on the calendar 4. Special Requirements The date and time provided by the network is needed. 5. Pre-conditions Empty or not updated calendar. 6. Post-conditions Updated calendar with content. 7. Extension Points Task creation page

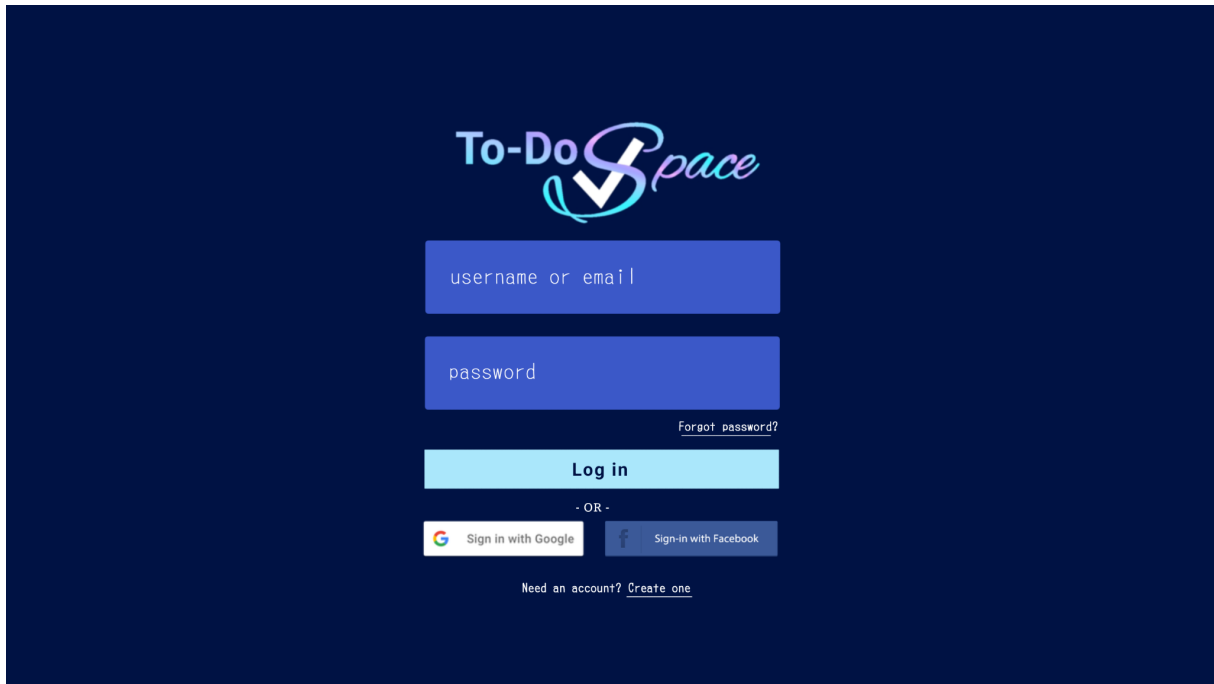
Create task	1. Brief Description Creates a task in its folder and relates it to all its linked items. 2. Actors User Other User 3. Flow of Events • <u>3.1. Main (basic) Flow</u> → The USER requests for the task creation page → The system provides the form to fill in to create the task → The user fills out the form: they add a Name to the task and may add other items (see alternate flow) → The USER submits the form → The system completes the task creation, sets creation date and default To-Do status → The system synchronizes the new task with the OTHER USERS who have access to the project the task is located in → The system displays the task on the project page → The system displays the task on the dashboard • <u>3.2. Alternate Flows</u> → When filling the form for task creation, the user wants to add a description, label, and attached file to the task → The system opens a text editor for the description and suggests advanced editing → The USER enables advanced editing and writes the description → The USER requests to add an attached file by clicking on the related button → The system redirects the user to the file manager of the computer → The USER chooses a file → The system downloads it → The system saves the description and attached file as linked content to the task → The USER requests to add a label by clicking on the related button → The system displays the label list → The USER chooses one or several labels → The system creates a shortcut to the task in the label's content → The USER submits the form → The system completes the task creation 4. Special Requirements Authorization from the computer to access local files needed, integrated text editor needs to be in place. 5. Pre-conditions Number of tasks: N-1 6. Post-conditions New existing task n°N. 7. Extension Points Task description text editor
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	Task status Label list
Control task status	1. Brief Description Allows the user to change a task's status between To-Do, Ongoing, Issue, Done, and eventual custom statuses. 2. Actors User Other user 3. Flow of Events • <u>3.1. Main (basic) Flow</u> → The USER requests to change the task status → The system suggests using the available statuses: To-Do, Ongoing, Issue, Done, and eventual custom statuses. → The USER chooses a new status for the task → The system synchronizes the new task status with the OTHER USERS who have access to the project the task is located in 4. Special Requirements / 5. Pre-conditions Task status X 6. Post-conditions Task status Y 7. Extension Points /
Display project	1. Brief Description Allows the user to display the project in various ways: as a table, kanban board, 3D view, idea cloud or an eventual custom view. 2. Actors User 3. Flow of Events • <u>3.1. Main (basic) Flow</u> → The system displays the project in a tabular overview by default. It contains lines and columns with categories and the tasks. • <u>3.2. Alternate Flows</u> 3.2.1 Kanban board → The USER changes the view to Kanban board by using the related button → The system displays the board, with one column per status and sorts the tasks accordingly.

	3.2.2 3D view → The USER changes the view to 3D view by using the related button → The system sets up X,Y and Z axis, Creates dedicated areas for different categories → The system compares the existing tasks with one another and determines their position on the 3D space → The system displays the task at their right position and enables manual rotation for the user → The USER can move freely in the 3D space 3.2.3 Idea cloud → The USER changes the view to Idea cloud by using the related button → The system compares the existing tasks with one another, takes categories into account, and determines their distance between each other → Using the calculated distance between each task and random the system shows the idea cloud by displaying each task one by one on the blank board 4. Special Requirements Integrated 3D software needed. 5. Pre-conditions Other or no displayed page 6. Post-conditions Project visible with chosen view type 7. Extension Points Folders Tasks Task statuses Create/Insert task Create/Insert label
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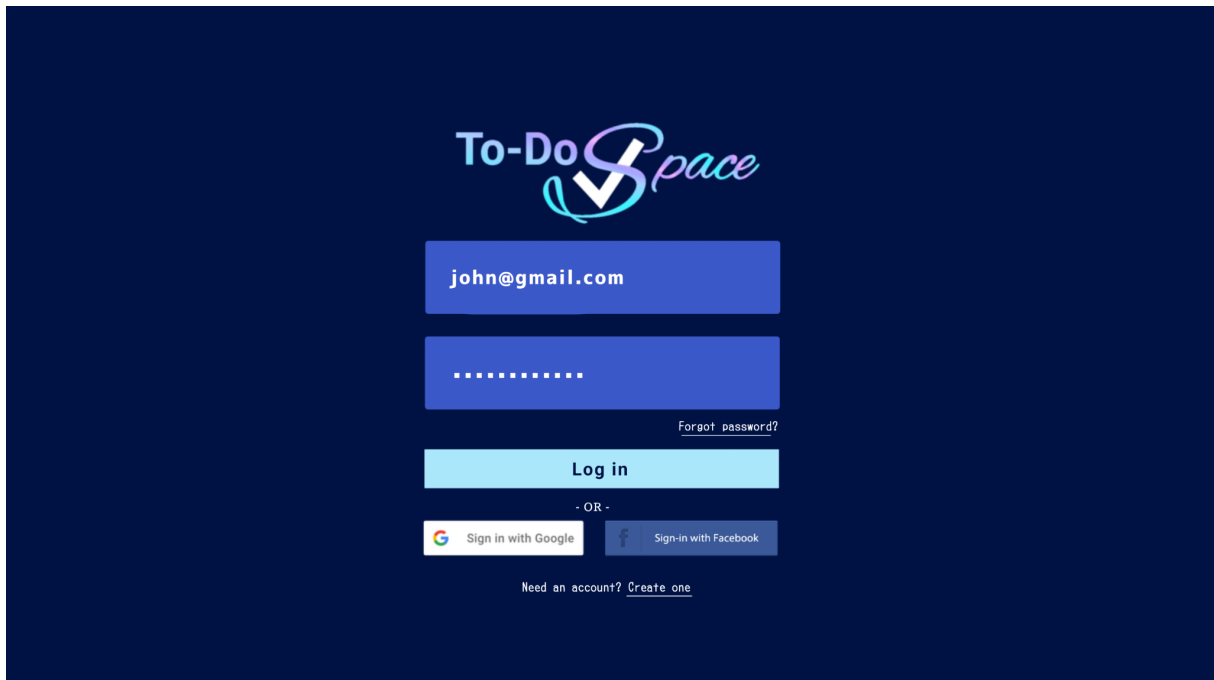
1.8 Prototype UI Views

Login Screen (no provided information):



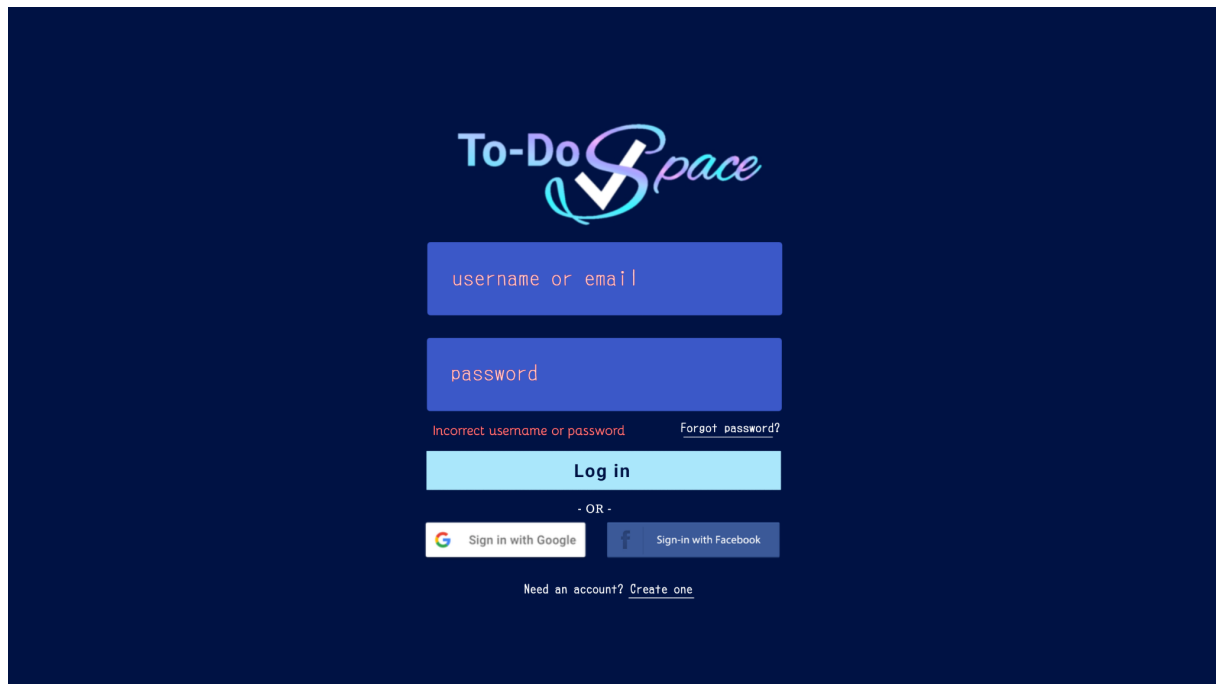
The login screen for To-Do Space features a dark blue background. At the top center is the logo, which reads "To-Do Space" in a stylized font with a checkmark integrated into the letter "o". Below the logo are two light blue input fields. The first field is labeled "username or email" and the second is labeled "password". To the right of the password field is a link that says "Forgot password?". Below these fields is a light blue button labeled "Log in". Underneath the button is the text "- OR -". Below this are two buttons: "Sign in with Google" (with the Google logo) and "Sign-in with Facebook" (with the Facebook logo). At the bottom center, there is a link that says "Need an account? Create one".

Login Screen (email and password provided but not submitted):

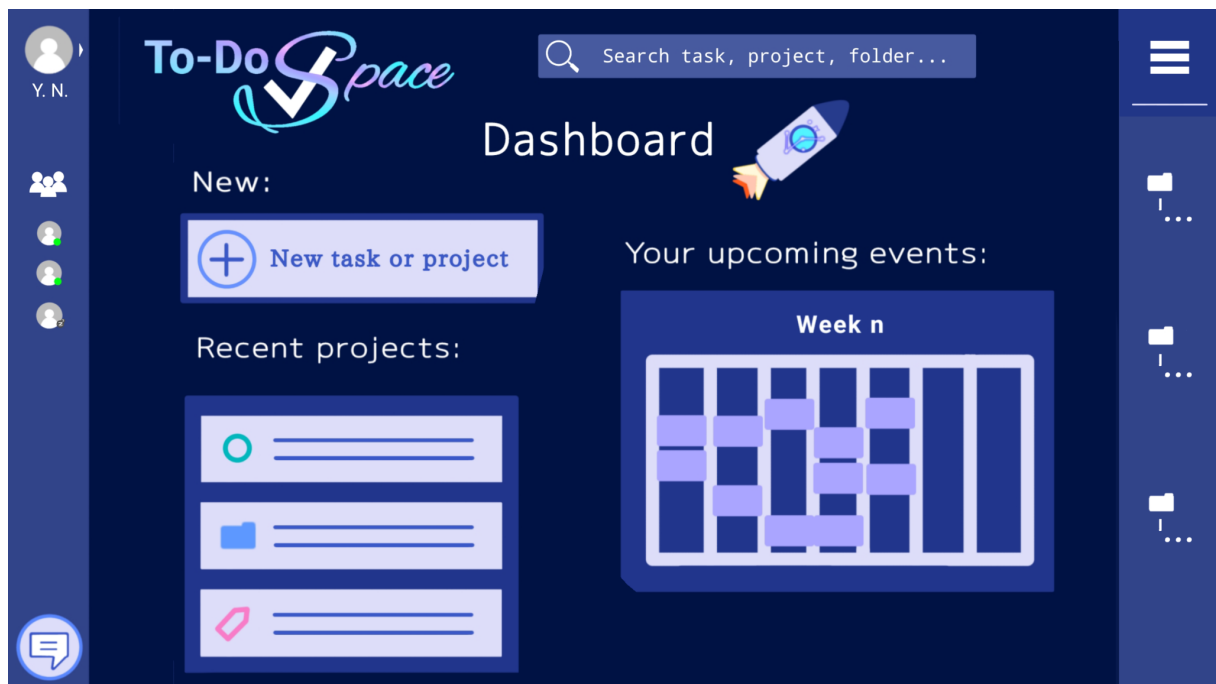


This login screen is identical in layout to the previous one, but the input fields contain information. The "username or email" field now contains the text "john@gmail.com". The "password" field contains ten dots, indicating a masked password. The "Forgot password?" link remains to the right of the password field. The "Log in" button, "- OR -" text, and the social media login buttons ("Sign in with Google" and "Sign-in with Facebook") are unchanged. The "Need an account? Create one" link is still at the bottom center.

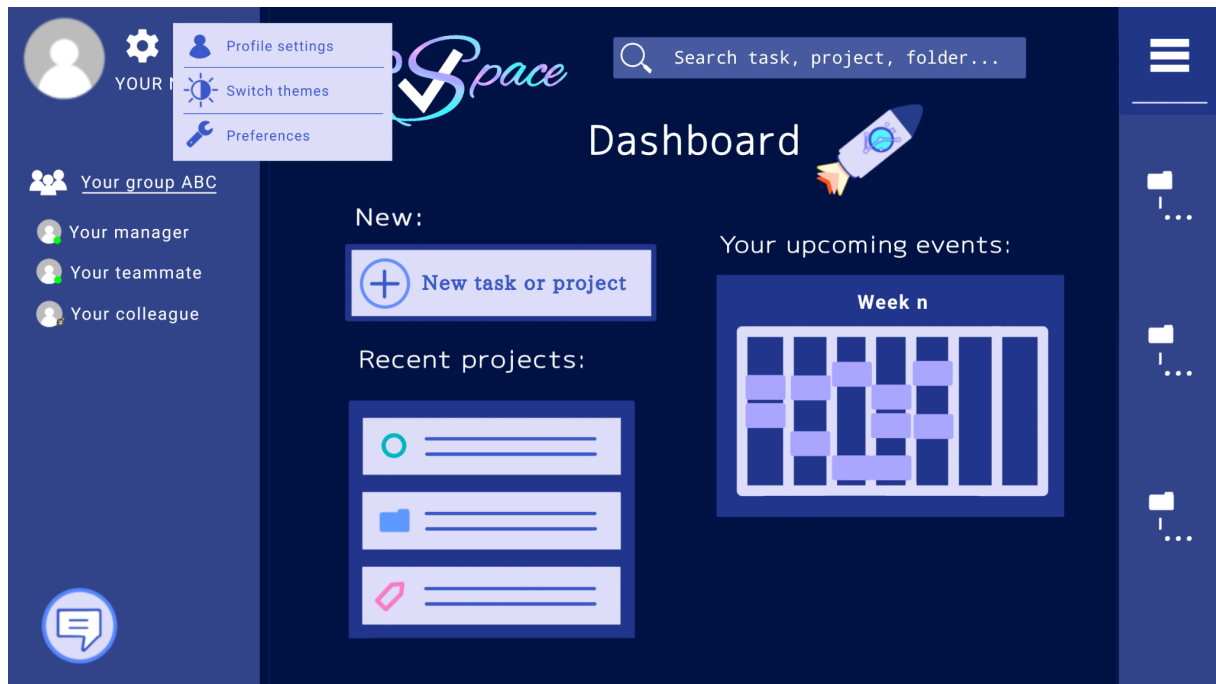
Login Screen (incorrect credentials):



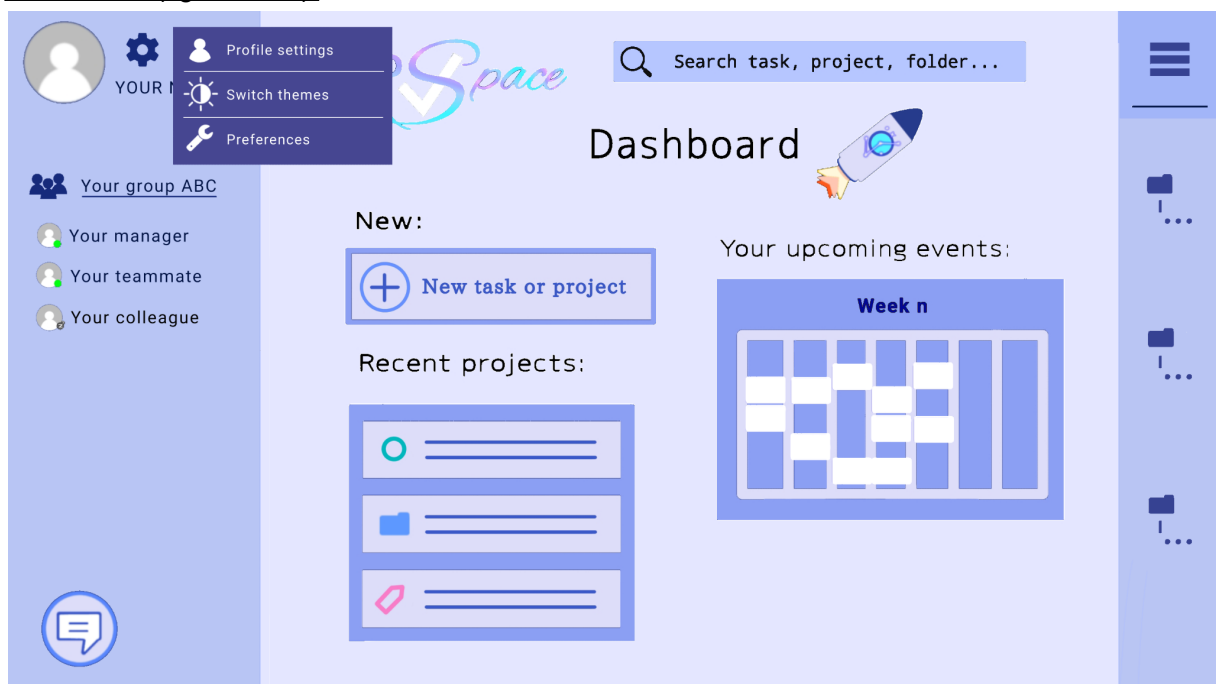
Dashboard (side tabs minimized):



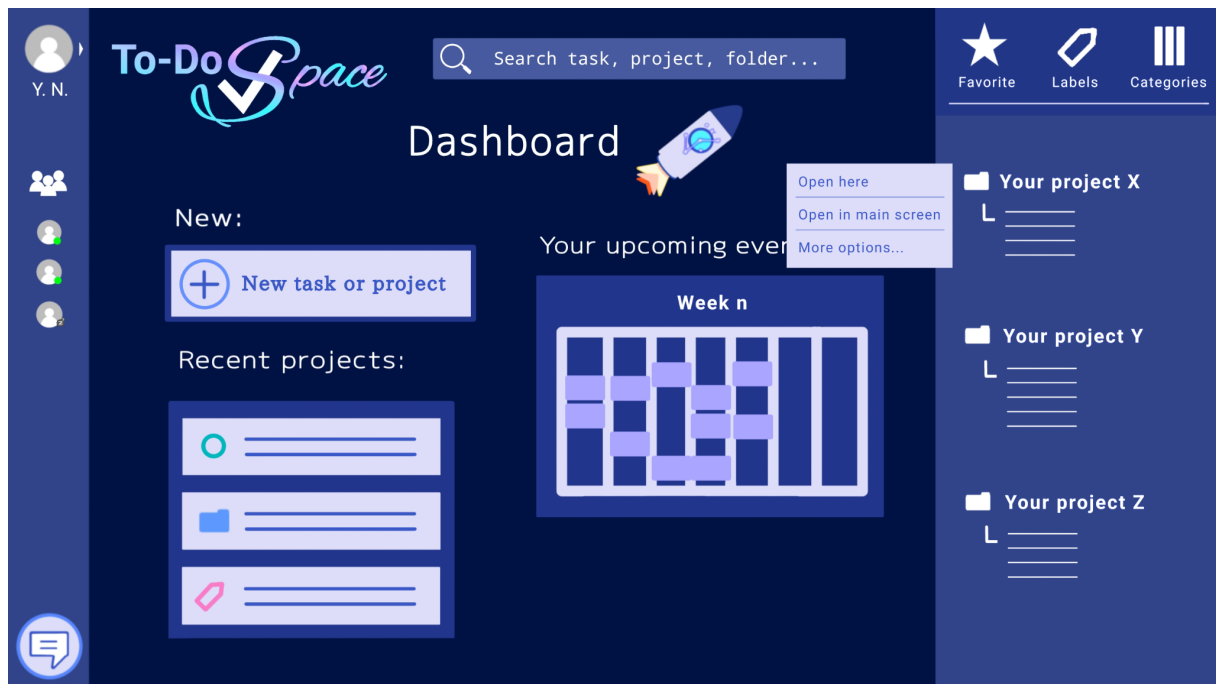
Dashboard (profile and user tab open, settings pop up):



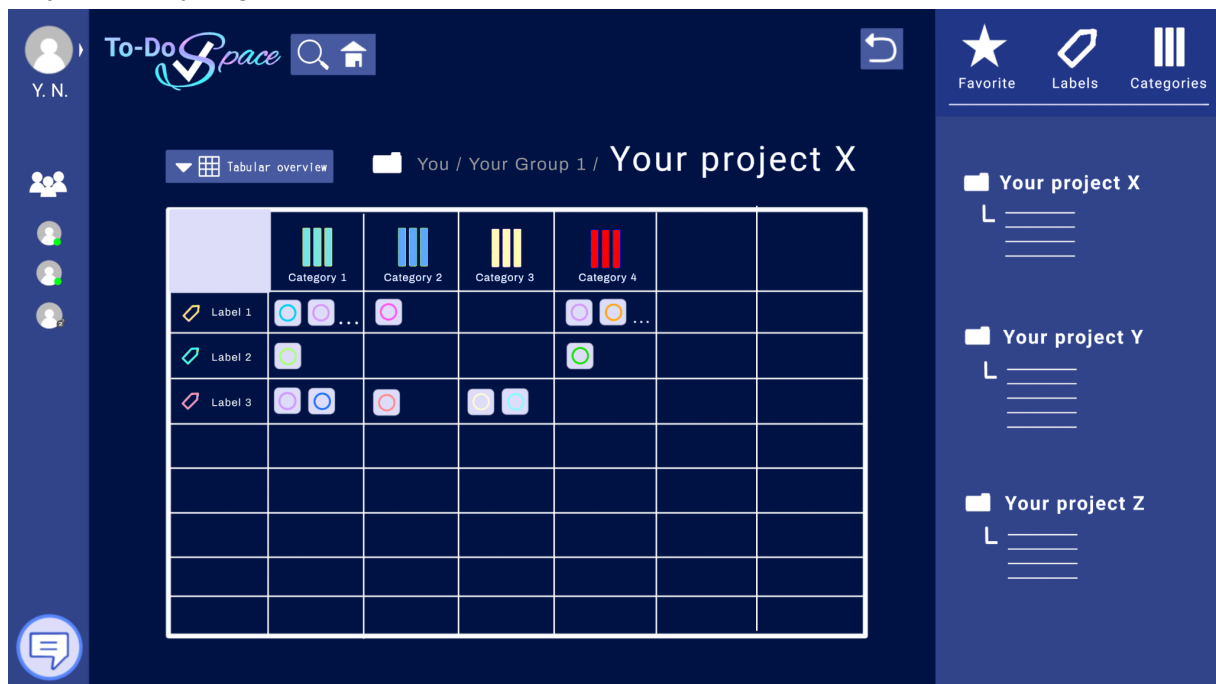
Dashboard (light mode):



Dashboard (project tab open with folder popup):



Project display page:



2. Detailed requirements

2.1. Functional requirements

-Account creation with Google / Facebook / Org. credentials:

Description: The user may create an account with one of the following methods: An e-mail + password, their Google credentials, their Facebook credentials, or the credentials that their institution has provided if the institution is linked with To Do Space. The method they have used when creating an account will remain the same for future logins.

Constraints: All users must have an account to access To Do Space.

Comments: Consider adding two factor authentication and/or a possibility for the user to log in with several methods.

-Personal information required for the account creation:

Description: Filling basic personal information, like a name, username and birthdate (for legal purposes), should be required to finalize the creation of an account.

Comments: 1. Information like a biography or interest fields tags may be filled in optionally.
2. Data such as the birthdate might be used for other features of the application, like the birthday reminder option on the calendar.

-Account management inside company: hierarchy

Description: The head of an organization or supervisor of a household may create groups (divisions and subdivisions) for the project and give progress monitoring permissions to the leaders of each division / subdivision.

Constraints: Although leaders should be able to monitor their group's work on the project, any employee with a To Do Space account should have the right to create other projects that are kept private if needed, for privacy reasons.

Comments: The progress may be monitored in various ways, for example by using percentages/stats, Kanban boards, reports, etc...

-Tutorial:

Description: The tutorial should appear after the installation of the app, showing the most important features to the user. If the user wants to use more complex features, the tutorial will continue by the moment he tries to use them.

As such, it is a way to find a balance between simplicity and functionality, since the most simple features will be at the foreground, easily accessible to users that seek simplicity, and complex features will still be available for those who seek functionality - but with a little bit of discovery required.

Comment: A "skip tutorial" option should be available to users who are already familiar with this kind of app.

-Dashboard:

Description: All of the user's ongoing tasks and projects should be visible on their dashboard, the home menu of the To Do Space. This is where they will be able to see the folders they created, the recently created tasks, the contact information of the group they belong to, ect...

-Items of the application: tasks, projects, folders, labels, custom items:

Description: Features that are included in the tasks and accounts excluded, the main items of the application should be the tasks, the projects and eventual additional folders inside the project, and the labels assigned to these items. These will serve as a foundation for the different types of display the app will offer.

Comment: The user may have the possibility to create custom items (by choosing a file type that is supported by the app and a custom icon) using the templates of the projects, folders and tasks, if he enables this option.

-Display options for the tasks/project:

Description: *Default display:*

Projects will be displayed as the main/root folders.

Categories will be displayed as columns or lines if a board / table display type is chosen, otherwise they will be displayed as labels. Tasks will be displayed as items within the categories, and will be able to carry several labels.

The location and status of the tasks will be displayed as labels.

Customization: The user should be free to display any items in the form they want it to be (for example, displaying the task statuses as folders). They should also be able to choose different view types such as a Kanban board, a table taking different parameters as columns and lines, an orthonormal coordinate system if there are two parameters, a 3D view if there are 3 parameters, a label/tag list, ect...

Comment: Display type templates should be created to fit the needs of different users without losing the most casual ones into the sea of infinite display possibilities. Companies or organizations should also be able to create their own display type templates and share them to their users.

-Tasks (characteristics):

Description: A task is characterized by a name, path (location in the folders), description, status, labels/tags creation and modification dates, and optional due date.

Constraints: To exist, a task must at least have a name that is unique within the folder it belongs to. Other fields can be left blank if needed.

Comments: There may be different task formats in use if we consider import/export options. Imported tasks of a different format than the default one may have less characteristics.

-Task “issue” status:

Description: In addition to the common “To-Do”, “Ongoing” and “Done” statuses provided by usual to-do applications, To Do Space provides an “issue” status for the tasks, that will automatically send a notification to (or display the contact info of) the project or group manager (or responsible person for technical help) when activated, in order to help the user fixing their problem as quickly as possible.

Constraints: The user may only get helped by members of his team and on open projects he belongs to, which are both defined by his organization. A global collaboration / help feature for all types of projects has not yet been considered as it would require to complexify the structure of the accounts and groups / networks.

Comment: Other custom statuses may be applied to the task if the user wants to.

-Progress tracking (with or without details, private/public view):

Description: A progress tracking feature will be available in different forms to satisfy the needs of different actors. For example, for an external actor, a rough estimation of the project progress through charts may be displayed, as confidentiality on the project content may be needed. For project managers on the other hand, a detailed view of the progress on each task may be required.

This feature may display the progress of a project or task using percentages, charts, colored indicators, remaining work time estimations, or a detailed view of completed / ongoing tasks.

-Task comment section (+emoji feedback):

Description: A comment section may be used for each task or project, where the user can report eventual issues, leave a note to another user, get feedback ect.

Comment: A quick feedback option, using a “like” or an emoji, is to be considered.

- Quicklinks to requirements/guidelines/references:

Description: A user should be able to easily reference another member of his group’s task into, using a quicklink.

Constraints: There are no limits to the kind of task the user can reference as long as they can see it (are a member of the reference’s project group / organization).

Comment: This is only applicable to tasks or projects that contain a description or an attached file, as there is no use in referencing an empty task.

-Attached files function:

Description: It should be possible to attach files to the description tasks.

Constraints: All file types should be supported, except for files that seem to be of a malicious type.

Comment: Depending on the account type a maximum file size to upload may be applied.

-Timetable:

Description: A timetable should be possible to add to the user’s dashboard in order to plan tasks accordingly.

-Calendar for tasks:

Description: For routines or periodically repeated tasks (for example; birthdays, scrum meetings), a calendar may be added to the dashboard. It will be in charge of automatically copying the right tasks at the right moment to the dashboard, following the cycle defined by the user.

Comment: The user should be able to manually stop this process if the routine ends.

-Task content transfer:

Description: Similarly to the calendar, if the task copier detects the creation of a new task with a name that is similar to a previous one (example: Grocery list // Shopping list. The elements that are common in all files having a name containing these words will get copied to the next one that is created).

Constraints: Contrarily to the calendar this duplication process should not be automatic and require the user's approval each time it is suggested to be effective (unless if the user manually sets this feature to automatic).

-Project resources management:

Description: For projects that involve the use of physical resources, the user should be able to track the remaining quantities of each resource, and get usage statistics in order to estimate how long the resources will last. Product suggestions based on the previously used resources, linked with product ratings, might be displayed when the remaining quantities are set to low.

Comment: Here product suggestions can be a source of revenue for the product owner.

-Task import and export:

Description: On the application, the tasks should be saved on a cloud using a specific format, and exportable to the computer using this and/or other formats. The exported tasks should be possible to import back again on any project. As such users would be able to share tasks or projects like tutorials freely as downloadable files on the internet.

Comment: there is a risk that the chosen task format may be used to propagate malware or malicious content on the internet. Prepare countermeasures to this issue during the design of the feature.

Task scan:

Description: An intelligent camera that is able to convert a photo of a simple series of instructions like a recipe or a tutorial to a Project or Task with subtasks should be available.

Comment: This artificial intelligence may require a lot of testing and learning during the phase following the application launch to be functional.

- Social media synchronisation:

Description: If a user links their To Do Space account to their social media account(s) (for example; twitter or facebook), they should be able to publish some of their tasks' content on the linked social media directly within the app.

Constraints: This feature should only be available to the owner or designated responsible party/ies of the project, to avoid any unwanted leaks.

-Intelligent tutorial or product suggestion:

Description: If the title of a task or project corresponds to something recognized by an AI (for example, if the title is "Bake pizza", or "Buy shampoo"), the AI should suggest related content that could help the user completing his task (here for example, a pizza recipe video, or a shampoo recommendation) directly on the app.

Constraints: To avoid giving irrelevant or even annoying suggestions to the user, the AI should only suggest widely reviewed pages (like popular youtube videos or wikiHow tutorials), at least during the first phase after the application launch. Data from the most used

suggestions during this phase could then be collected and used to improve the recommendation algorithm. This feature should also be possible to disable manually by the user.

Comment: Product suggestion can also be used as a way to generate revenue to the product owner.

-Export task to text / report format:

Description: Tasks that are labeled as done should be easily exportable to a visual report template in a text format, so they can be added to the product backlog or sprint backlog if needed.

This template should, by default, show the characteristics of the task (project, category, labels, name, description...), the time spent on it, compared to the estimated time if given (+ additional statistics), the unexpected issues that were faced (if encountered), the comment section of the task (if used), and the final feedback.

Comment: Of course, the user can freely modify this given template in order to get the right information for their product and sprint backlogs.

-Integrated search engine and automatised bibliography:

Description: For some specific tasks like the writing of a text or scientific essay for example, the user should have access to a full text editor linked to the task as the task content/description (or the display of a connected personal text editor like Word, Overleaf, Google docs...). Alongside this text editor should the application be able to display a picture-in-picture integrated search engine (like DuckDuckGo, Google or Ecosia for general needs, or ScienceDirect for scientific needs for example). This combination of an integrated search engine and text editor would then automatise the bibliography, since the workspace directly gathers the references of the publications the author is using. As such, this would provide an all-in-one project workspace for authors and members of the scientific community (hence the name To-Do Space).

-Eco-friendly features (to be considered):

Description: As mentioned in the project resources management, intelligent product suggestions and integrated search engine feature descriptions, external algorithms should support the application to provide these features. This is an opportunity to choose eco-friendly building blocks for these features: like setting Ecosia as a default general search engine, or suggesting environment-friendly products for example.

2.2. Non-functional requirements

2.2.1 Security requirements:

Category: Security

Name: **2 Factor Authentication**

Description: The default authentication method should be a regular username and password, or a login via a supported platform (Facebook, Google or a company's platform).

However, if the user wants to enable it, they should be able to have 2 factor authentication on their account. This change should be available through the settings menu.

This second factor should be an SMS or email authentication code, or a verification through a dedicated app like Google authenticator for example.

Acceptance criteria (according to Hermes 5): Importance: 3 Urgency: 2 Risk: 2 Outlay: 2

—

Category: Security

Name: **Password strength**

Description: The user's password should reach the following level of strength to be accepted during registration or change:

- A minimum of 12 characters

- Variations between the use of capital and lowercase letters.

- At least one non-letter character (can be a digit or any punctuation mark).

- Not more than 3 digits in a row (in order to avoid the use of birthdates).

Note: All types of digits available with UTF-32 (except for control commands like backspace for example) should be available for the creation of the password.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 3 Risk: 3 (depending on whether it is possible or not to make all characters types available) Outlay: 2

—

Category: Security

Name: **Authentication attempts and captcha puzzle**

Description: All authentication attempts should be registered. If an authentication remains unsuccessful after 10 attempts (except when the error is network related), the connection on

this account should be blocked and the user should receive an e-mail or an SMS asking them to reset their password if they want to continue.

During registration, password change and every once in a while for the login, a captcha puzzle should be completed by the user for them to access the following page.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 3 Risk: 2 Outlay: 2.5

—

Category: Security

Name: **Encryption**

Description: The data that a user enters should be encrypted with a strong encryption algorithm, preferably AES256, when migrating to the server (or otherwise, another algorithm that is compatible with the building tools and components of the app). The passwords should use SHA512 encryption. All personal data stored on the servers should be kept private and secure from potential attacks coming from outside.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 4 Risk: 2 Outlay: 2.5

—

Category: Security

Name: **Access and control**

Description: The integrated services coming from other applications or websites should be stored on the database, with their corresponding authorization levels. All other third party apps that try to access some of the user's data should be blocked.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 4 Risk: 2 Outlay: 2

2.2.2 Interoperability requirements:

Category: Interoperability

Name: **Platform format**

Description: The application should be displayed in a regular HTML format, in order to be available on any laptop web browser regardless of the device. Compatibility with smartphones or other device types is to be considered.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 4 Risk: 2 Outlay: 3

2.2.3 Performance requirements:

Category: Performance

Name: **Latency**

Description: Changes in the application should occur at a reasonable speed, optimally at 30 ms plus the required time for the changes on server scale. If the response time is more than 300ms, switch to offline mode. Data will then be stored on the user's device instead and then be uploaded whenever the connection to the server gets recovered.

See KPIs for details.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 4 Risk: 1 Outlay: 2

—

Category: Performance

Name: **Throughput**

Description: There should be no limit on the traffic speed when the application is used by a small number of users simultaneously. If a higher number of users connect simultaneously, adjust the maximum traffic speed accordingly. This limit is to be determined.

If the throughput is less than 10kb/s, switch to offline mode. Data will then be stored on the user's device instead and then be uploaded whenever the connection to the server gets recovered.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 4 Risk: 1 Outlay: 2

KPIs:

Optimal latency: **<35ms**

Latency that is considered reasonable: **< 300ms**

Optimal throughput: **>10 mb/s**

Throughput that is considered reasonable: **< 100 kb/s**

Switch to offline mode: **Latency > 300ms, throughput > 100 kb/s**

Query processing time (read only): **1 second**

Query processing time (write): **2 seconds**

Session duration: **expect an average 20 minutes active, 4 hours in the background**

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Session expiration: **Set by user, 2 hours of inactivity by default**

2.2.4 Availability:

Category: Availability

Name: **Uptime**

Description: The server should be available 99.995% of the time, optimally.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 5 Risk: 2 Outlay: 2

—

Category: Availability

Name: **Q&A and Support team**

Description: Support team verifications for critical problems should be done daily. Non-critical issues such as customer support should be examined weekly, with a Q&A forum on the use of the app available 24/7

Acceptance criteria (according to Hermes 5): Importance: 4 Urgency: 3 Risk: 2 Outlay: 3

—

Category: Availability

Name: **Offline mode**

Description: If the servers become unavailable during maintenance, or the connection is lost, data will then be stored on the user's device instead and then be uploaded whenever the connection to the server gets recovered. All personal projects will still be available however group work changes won't be applied immediately, and functionalities such as the chat will be suspended.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 3 Risk: 2 Outlay: 2

KPIs:

Availability requirements for different geographies: **Not available in countries that do not support the linked applications (ex. Google in china)**

Frequency of scheduled planned outage activity that is acceptable: **2-3 hours every 3 months**

Expected unplanned outage activity: **25-30 hours during the launch year**

Expected overall expected server uptime: **99.995%**

Service Desk Availability (for non-critical situations): **Q&As open 24/7, support available weekly**

Support team availability (for critical situations): **Daily**

2.2.5 Scalability requirements:

Category: Scalability

Name: **Global data storage**

Description: The global storage provided at launch should be set to 5TB, allowing each user to store 5GB of data on a free version, thus supporting 1000 users. More storage should be available with a premium model to elaborate, including more advanced features. Depending on the demand after launch, increase the storage size accordingly.

Acceptance criteria (according to Hermes 5): Importance: 5 Urgency: 4 Risk: 3 Outlay: 1

—

Category: Scalability

Name: **User data limits**

Description: A free user should be able to store up to 5GB of data on the cloud, and a total of 1000 tasks. Tasks added in shared projects are considered to be those of the project owner.

Acceptance criteria (according to Hermes 5): Importance: 4 Urgency: 2 Risk: 1 Outlay: 2

KPIs:

Ability to scale up to n active users in a year: **To be determined, expect 10 000 users at launch**

Number of concurrent users during peak periods: **TBD, expect 500 users at launch**

Session duration: **expect an average 20 minutes active, 4 hours in the background**

Query processing time (read only, open page): **1 second**

Query processing time (write): **2 seconds**

Number of tasks per free user: **1000**

Data quantity stored per free user: **5GB**

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